Effects of Inlet Diameter and Sac Asymmetry on Transitional Flow in Idealized Descending Thoracic Aortic Aneurysms, with Neural-Field Recovery of Pressure and Wall Shear Stress from Sparse Velocity

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Abstract

We examined how inlet diameter and sac asymmetry affect flow in twelve idealized descending thoracic aortic aneurysm (DTAA) geometries, then used their computational fluid dynamics (CFD) fields to test pressure and wall shear stress (WSS) recovery from sparse velocity. Nine aneurysmal models combined three inlet diameters (2.0, 2.3 and 2.6 cm) with axisymmetric, anterior-dominant and posterior-dominant sacs; three models served as controls. All aneurysms had a maximum diameter of 4.0 cm, and all vessels had an outlet-to-inlet diameter ratio of 0.8. At a fixed flow waveform, narrowing the inlet from 2.6 to 2.0 cm increased systolic pressure drop from 528–561 to 1261–1346 Pa and the 95th-percentile WSS from 7–8 to 15–18 Pa. Asymmetry mainly changed the location of low shear and increased diastolic segment-median WSS by a factor of 1.8–3.1 relative to axisymmetric sacs. Reconstruction used noise-free velocity sampled at about 1.3% of interior nodes at two instants 5–10 ms apart. Interpolation and space-time neural fields were compared using shared pressure and shear calculations. In seven evaluation geometries, the continuity-constrained field reduced pressure and WSS errors relative to the prespecified radial basis function (RBF) in all 14 case–phase means, with median error ratios of 0.46 and 0.51. Most of the benefit came from the neural representation; momentum-constrained physics-informed neural networks (PINNs) tested on two development geometries did not improve joint pressure and shear recovery. Diastolic pressure estimates depended on CFD-supplied eddy viscosity, and WSS accuracy deteriorated when near-wall observations were removed. These results characterize reconstruction in known, idealized geometries; they do not establish performance on measured data or clinical rupture risk.

Keywords. Thoracic aortic aneurysm; computational fluid dynamics; wall shear stress; sparse flow reconstruction; physics-informed neural networks; relative pressure estimation

1 Introduction

Descending thoracic aortic aneurysms alter both blood flow and the mechanical loading of the vessel wall. Their geometry varies in parent-vessel size, curvature and sac asymmetry. These features can differ even when two aneurysms have the same maximum diameter, motivating comparisons that control diameter while varying the surrounding geometry [1, 2].

Maximum diameter is an important clinical measure, but it does not describe wall thickness, tissue properties or the full aneurysm shape. Evidence from abdominal aortic aneurysms shows that diameter alone does not determine rupture [3] and has motivated complementary volumetric measures [4]. These observations motivate biomechanical analysis, but findings from abdominal or cerebral aneurysms cannot by themselves establish rupture predictors for descending thoracic aortic aneurysm (DTAA). Structural tensile stress and hemodynamic wall shear stress (WSS) also describe different loads: local wall failure depends on the relation between tissue stress and tissue strength, whereas WSS characterizes tangential loading by blood flow [2, 5].

Computational fluid dynamics (CFD) estimates velocity, pressure and shear fields that are difficult to measure throughout an aneurysm [6, 7]. Common descriptors include WSS, wall shear stress gradient (WSSG), time-averaged wall shear stress (TAWSS) and oscillatory shear index (OSI). Their associations with disease depend on the anatomy and outcome studied. Geers et al. [8], for example, examined shear at cerebral aneurysm initiation sites, whereas Moon et al. [9] reported low TAWSS and elevated oscillatory and residence-time indices at rupture regions in four infrarenal aneurysms. Such findings motivate spatial hemodynamic comparisons without establishing a validated rupture model for the idealized thoracic geometries studied here.

Parent-vessel diameter and curvature affect the jet entering an aneurysm and the resulting recirculation. Studies of tortuous arteries have linked these geometric features to changes in shear and residence-time indices [10–14]. In intracranial aneurysm models, larger parent-vessel diameters reduced WSS and OSI [15, 16], while changes in aspect and size ratios altered near-neck hemodynamics [17]. A controlled thoracic-aorta model allows these geometric effects to be examined without the simultaneous anatomical variation present in patient cohorts.

Pulsatility, curvature and sac expansion can produce transitional flow, making the choice of flow model relevant to the calculated hemodynamics [18–21]. Reynolds-averaged turbulence models represent the effects of unresolved fluctuations through a closure; direct numerical simulation (DNS) instead resolves the turbulent scales directly. Here we use the transition shear stress transport (SST) k–ω model to represent possible laminar–turbulent transition. The resulting turbulence quantities are model predictions and require validation before being interpreted as measurements of physiological turbulence.

Several studies have investigated aneurysm hemodynamics using patient-based and idealized geometries. Lauric et al. [22] examined 33 middle cerebral artery aneurysm patients and 44 controls, finding aneurysmal arteries have a tapered parent vessel (mean ratio 0.88) versus near-unity in healthy vessels; tapering elevates WSS and WSSG, promoting aneurysm formation. Jin et al. [23] reported that reducing coil porosity from 0.89 to 0.79 decreases OSI by up to 55% in an internal carotid artery (ICA) aneurysm. Tian et al. [24] used 18 idealized and 7 patient-specific bifurcation aneurysms to show that the aneurysm–parent ves-

sel angle dominates low-density lipoprotein (LDL) deposition, with larger angles promoting deposition by reducing inflow and enhancing stagnation. Rehman et al. [25] studied Marfan syndrome aneurysms using fluid–structure interaction (FSI) with CFD and physics-informed neural networks (PINNs) on six patient-specific models, observing that WSS disturbances vary with shape and that the anterior wall experiences higher stresses. Zhou et al. [26] developed an AutoML model on 158 intracranial aneurysms, identifying bifurcation angle as the most critical risk factor. These patient-based studies, however, are confounded by multiple simultaneously varying anatomical features, making it difficult to isolate single-parameter effects. Idealized geometries offer systematic parameter variation.

The present study examines a curved, tapered family of twelve idealized descending thoracic aortas (Table 1). Nine aneurysmal geometries combine inlet diameters of 2.0, 2.3 and 2.6 cm with axisymmetric, anterior-dominant and posterior-dominant sacs; three nonaneurysmal geometries provide matched controls. The maximum sac diameter is fixed at 4.0 cm, and every vessel has an outlet-to-inlet diameter ratio of 0.8. This design compares inlet diameter and bulge direction under a common taper and flow waveform. Because taper is fixed, the study does not estimate the effect of changing taper itself.

The same CFD fields support a second question: how accurately can pressure and WSS be recovered from sparse velocity in a known geometry? Time-resolved three-dimensional phase-contrast MRI (4D flow MRI) samples velocity on voxels of roughly one to three millimeters [27]; pressure and wall shear must be estimated from those measurements. Relative-pressure methods use the momentum balance [28,29], while velocity-derived WSS is sensitive to resolution and wall segmentation [30–32]. Neural fields provide a continuous velocity representation, optionally constrained by the governing equations [33,34]. Applications include arterial pressure inference [35], 4D flow reconstruction [36,37], near-wall flow recovery [38] and aneurysm flow reconstruction [39]. Their performance depends on the observations, training procedure and comparator [40–42].

We separate the benefit of the neural representation from that of the physics terms in its loss. We also test which additional inputs pressure and shear recovery require. In unsteady Reynolds-averaged flow, the pressure gradient depends on acceleration and the eddy-viscosity closure as well as velocity. All methods therefore use the same sampled observation pool and shared pressure and shear calculations, with closure information reported explicitly. Table 1 places this comparison alongside earlier idealized-aneurysm studies.

This study addresses both questions on a family of twelve idealized descending thoracic aortic geometries, namely nine aneurysmal models formed by three inlet diameters and three bulge patterns at a fixed 4.0 cm maximum sac diameter, and three nonaneurysmal controls. The contributions are as follows.

1. We report how inlet diameter and sac asymmetry change pressure drop, reversed flow, eddy viscosity and aneurysm-zone WSS in transient transitional CFD, evaluated at common systolic and diastolic instants for all twelve geometries.

2. We define a sparse-reconstruction task on the same fields, in which velocity is observed at about 1.3% of the interior nodes at two instants 5–10 ms apart and hidden pressure and aneurysm-zone WSS must be recovered in the known geometry. Linear interpolation, radial basis function (RBF) interpolation and space-time neural fields with and without a continuity penalty all pass through the same momentum-integration and

Table 1: Comparison of existing idealized aneurysm studies and the present work, emphasizing differences in geometry, flow modeling, boundary conditions, hemodynamic–morphometric analysis, and sparse-velocity reconstruction.

Feature	**Jin et al. [43]**	**Guerrero-Hurtado et al. [44]**	**Rehman et al. [45]**	**Rehman et al. [46]**	**Moroni & Espa [47]**	**Present Study**
Year	2021	2023	2024	2025	2025	-
Geometry Type	Idealized	Idealized	Idealized	Idealized	Idealized	Idealized
DTAA	No	No	Yes	No	No	Yes
Arterial Tapering	No	No	No	No	No	Yes
Parent vessel effect	No	No	No	No	No	Yes
Morphology: Axisymmetric or Asymmetric	Not varied	Not varied	Both	Both	Both	Both
SST $k-\omega$ Transition Turbulence Model	No	No	No	No	No	Yes
Transient Flow Inlet	Yes	Yes	No	No	No	Yes
Transient Pressure Outlet	Yes	No	No	No	No	Yes
Plane-wise flow–morphometry analysis	Partial	No	No	Partial	Partial	Yes
PINN	No	No	Yes	No	No	Tested on two cases
Hidden pressure and WSS from sparse velocity against matched baselines	No	No	No	No	No	Yes

near-wall shear calculations, so their differences are attributable to the reconstructed velocity alone.

3. Using a protocol fixed on two development geometries before evaluation on seven others, we separate the contributions of the neural representation, the continuity constraint, the local time derivative and the eddy-viscosity closure. On two development geometries, we also report that the pressure output of a PINN with the momentum residual in its loss was far less accurate than momentum integration applied after fitting.

The geometry study supplies reference flows with different jet strengths, recirculation patterns and modeled eddy viscosities. The reconstruction study tests recovery across those conditions. Each reconstruction receives observations in its target geometry; it does not predict flow in an anatomy without observations or estimate clinical rupture risk.

2 Materials and methods

2.1 Geometric description

Twelve idealized descending-aortic geometries were examined, namely nine aneurysmal models and three nonaneurysmal (healthy) controls. The aneurysmal group used three inlet diameters, 2.0, 2.3 and 2.6 cm. An axisymmetric and two asymmetric aneurysm shapes were studied for each inlet diameter. The level of asymmetry was calculated as the ratio $\beta = r/R$,

where r and R denote the characteristic radii of the aneurysmal segment in the posterior and anterior directions, respectively. An axisymmetric bulge gives $\beta = 1$; a posterior-dominant bulge gives $\beta > 1$ and an anterior-dominant bulge $\beta < 1$, the two representing opposite directional dilations. The detailed geometric parameters are given in Table 2. To compare with the healthy models, the other three models that have the same inlet diameters (2.0 cm, 2.3 cm and 2.6 cm) were also built. In these control cases no aneurysmal dilation was added and, as in the aneurysmal models, the vessels were tapered smoothly along their lengths, with a physiologically consistent reduction in diameter so that the outlet diameter is 80% of the inlet diameter. The twelve cases therefore form three groups by inlet diameter, each containing three aneurysmal cases and their matching control. Group 1 (2.0 cm) comprises Cases 1–3 and control Case 10, Group 2 (2.3 cm) comprises Cases 4–6 and control Case 11, and Group 3 (2.6 cm) comprises Cases 7–9 and control Case 12. Case numbers are used throughout, and each control is compared with the aneurysmal cases of its own group.

The eight cross-sectional planes (D1–D8) shown in Figure 1 were used to investigate the spatial variations of the hemodynamic parameters along the arterial axis. All the aneurysmal models had a single dilation centered at the mid-region (D5) and the maximum aneurysm diameter was set at 4.0 cm for all the diseased aneurysm cases. The (D1–D4) and (D6–D8) represent the upstream and downstream portions of the aneurysms. The average diameters of the planes are listed in Table 3. Two section planes are used throughout, the medial plane (xy-plane) and the transverse plane (xz-plane) through the axis of the descending segment. Every model is mirror-symmetric about the medial plane, and the anterior–posterior asymmetry of the bulge lies within that plane.

Table 2: Aortic aneurysm (AA) models considered in the present study. Here r and R are the posterior and anterior characteristic radii of the sac and $\beta = r/R$ is the directional asymmetry ratio. Each group collects the three aneurysmal cases of one inlet diameter and their control.

		Sac geometry			Case ID by inlet diameter D_{in}		
Type	Configuration	r (cm)	R (cm)	$\beta = r/R$	Group 1 2.0 cm	Group 2 2.3 cm	Group 3 2.6 cm
Axisymmetric	–	2.0	2.0	1.00	1	4	7
Asymmetric	Ant.-dom.	1.3	2.7	0.48	2	5	8
	Post.-dom.	2.7	1.3	2.08	3	6	9
Control	–	–	–	–	10	11	12

2.2 Blood flow modeling and boundary conditions

2.2.1 Governing equations and turbulence modeling

The incompressible Navier–Stokes equations, which govern blood flow, are given by [48, 49]:

$$\nabla \cdot \mathbf{v} = 0, \tag{2.1}$$

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} \right) = -\nabla p + \mu \nabla^2 \mathbf{v}, \tag{2.2}$$

Table 3: Plane diameter (cm), mean ± standard deviation across the nine aneurysmal and three control geometries.

Plane	Aneurysmal ($n = 9$)	Control ($n = 3$)
D1	2.28 ± 0.26	2.28 ± 0.30
D2	2.22 ± 0.25	2.22 ± 0.29
D3	2.15 ± 0.24	2.15 ± 0.28
D4	2.97 ± 0.09	2.11 ± 0.28
D5	4.00 ± 0.00	2.08 ± 0.27
D6	2.92 ± 0.09	2.05 ± 0.27
D7	2.02 ± 0.23	2.02 ± 0.26
D8	1.95 ± 0.22	1.95 ± 0.25

where ρ is the fluid density, $\mathbf{v}$ the velocity vector, p the pressure, and μ the dynamic viscosity.

For the unsteady Reynolds-averaged calculation, the instantaneous velocity will be decomposed as $\mathbf{v} = \mathbf{V} + \mathbf{v}'$ with $\overline{\mathbf{v}'} = 0$, where $\mathbf{V}$ is the mean velocity. The Reynolds stress is closed by the Boussinesq eddy-viscosity approximation,

$$-\rho\overline{v_i' v_j'} = \mu_t \left(\partial_j V_i + \partial_i V_j\right) - \frac{2}{3}\rho k \delta_{ij}, \tag{2.3}$$

where $k = \frac{1}{2}\overline{v_i' v_i'}$ is turbulent kinetic energy (TKE) per unit mass ($\mathrm{m}^2\,\mathrm{s}^{-2}$), μ_t is dynamic eddy viscosity, and δ_{ij} is the Kronecker delta. With Eq. (2.3), the Reynolds-averaged momentum equation solved for the mean velocity and pressure P is

$$\rho\left(\frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V}\cdot\nabla)\mathbf{V}\right) = -\nabla P + \nabla\cdot\left[(\mu+\mu_t)\left(\nabla\mathbf{V} + \nabla\mathbf{V}^T\right)\right] - \frac{2}{3}\nabla(\rho k), \tag{2.4}$$

with $\nabla\cdot\mathbf{V} = 0$.

2.2.2 SST k–ω transition model

The SST k–ω transition model was employed to account for the possible transition from laminar to turbulent flow and to capture the complex flow features associated with curvature, flow separation, and recirculation in the descending thoracic aorta. The model combines the near-wall formulation of the k–ω model with the free-stream behavior of the k–ϵ formulation through appropriate blending functions.

The turbulent eddy viscosity is calculated as

$$\mu_t = \frac{a_1 \rho k}{\max(a_1\omega,\ SF_2)}, \tag{2.5}$$

where k is the TKE, ω is the specific dissipation rate, S is the magnitude of the strain-rate tensor, and F_2 is the second blending function.

The transport equations for TKE and specific dissipation rate are given by

$$\frac{\partial(\rho k)}{\partial t} + \nabla\cdot(\rho k\mathbf{V}) = \nabla\cdot\left[\left(\mu + \frac{\mu_t}{\sigma_k}\right)\nabla k\right] + P_k - D_k, \tag{2.6}$$

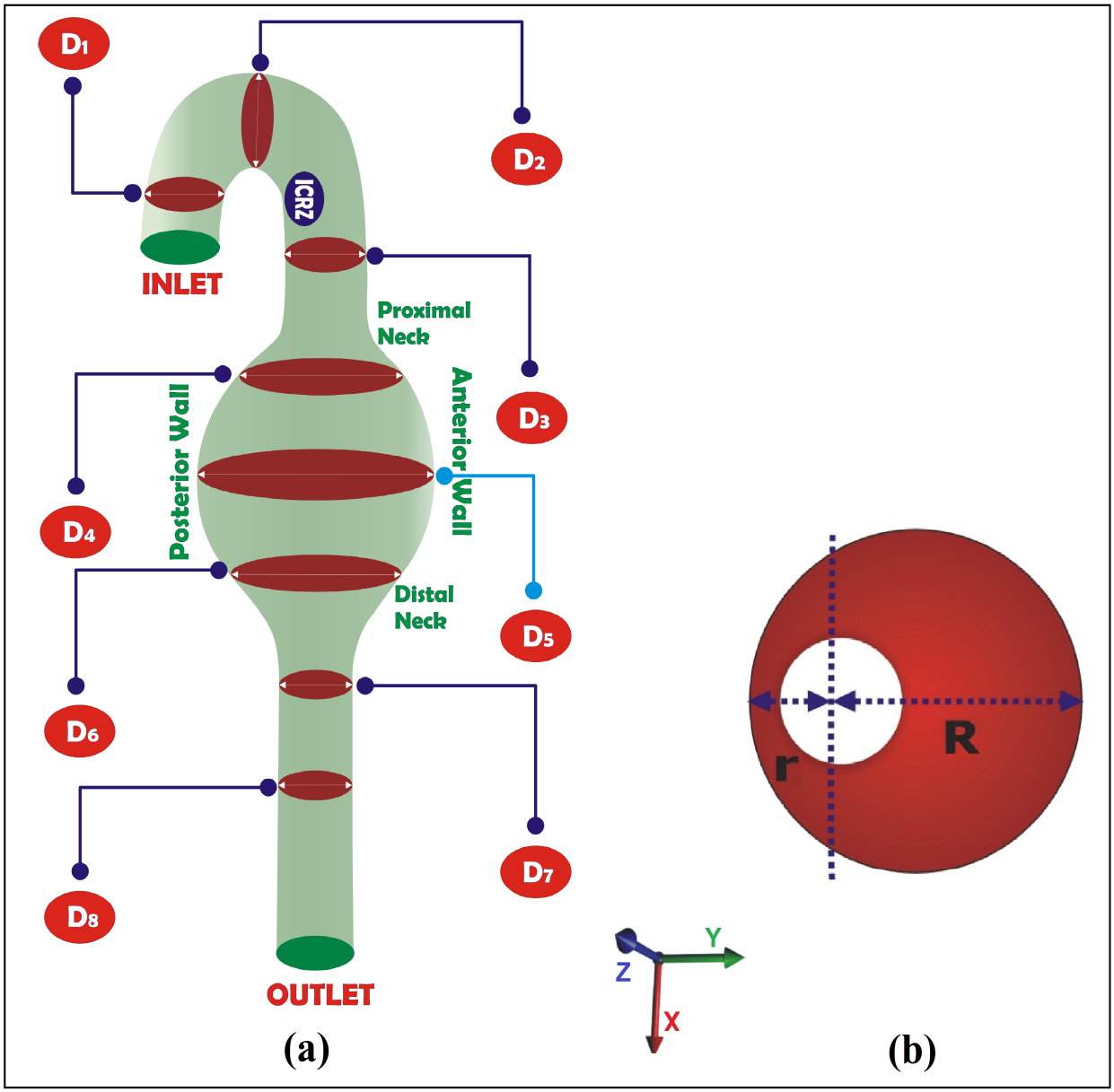


Figure 1: Visual depictions of (a) the three-dimensional geometry of the aneurysm and (b) the cross-sectional area of the aneurysm sac.

$$\frac{\partial(\rho\omega)}{\partial t} + \nabla\cdot(\rho\omega\mathbf{V}) = \nabla\cdot\left[\left(\mu + \frac{\mu_t}{\sigma_\omega}\right)\nabla\omega\right] + P_\omega - D_\omega + 2(1-F_1)\rho\sigma_{\omega 2}\frac{1}{\omega}\frac{\partial k}{\partial x_j}\frac{\partial \omega}{\partial x_j}. \quad (2.7)$$

Unlike the standard SST k–ω model, the transition formulation additionally solves two transport equations: one for the intermittency γ, which varies from zero in laminar regions to unity in fully turbulent regions, and another for the transition momentum-thickness Reynolds number $\widetilde{Re}_\theta$. These variables enable the model to predict the onset and development of boundary-layer transition based on the local flow conditions.

The blending functions F_1 and F_2 regulate the transition between the k–ω and k–ϵ

formulations. F_1 provides the appropriate near-wall blending, whereas F_2 limits the eddy viscosity in regions where strain-rate effects become important. The distance to the nearest wall is denoted by y. Thus, the SST k–ω transition model combines the strong near-wall behavior of the k–ω formulation with the reduced free-stream sensitivity of the k–ϵ formulation in the outer region of the boundary layer, while the additional intermittency and transition-Reynolds-number equations allow laminar–turbulent transition to be represented.

2.2.3 Boundary conditions

The pulsatile flow rate at the inlet is used in order to replicate the pulsatile nature of the blood flow in the human body. The inlet profile is expressed as:

$$Q(t) = \begin{cases} Q_0 \sin\left(\dfrac{\pi\,(t - nT)}{0.35}\right) + 0.05\,Q_0, & nT \le t \le nT + 0.35, \\ 0.05\,Q_0, & nT + 0.35 < t \le (n+1)T. \end{cases} \tag{2.8}$$

Similarly, the pulsatile outlet pressure is used at the outlet and is

$$P(t) = \begin{cases} P_{\min} + (P_{\max} - P_{\min}) \sin\left(\dfrac{\pi\,(t - n\,T)}{0.35}\right), & nT \le t \le nT + 0.35, \\ P_{\min}, & nT + 0.35 < t \le (n+1)T. \end{cases} \tag{2.9}$$

Here $P_{\min} = 80$ mmHg, $P_{\max} = 120$ mmHg, $n = 0, 1, 2, 3$ indexes cardiac cycles, $T = 0.8$ s, and $Q_0 = 2.39 \times 10^{-4}$ m^3 s^{-1} is the sinusoidal amplitude. Because of the $0.05Q_0$ offset, the actual peak of the prescribed inlet flow rate is $1.05Q_0$.

The inlet flow-rate and outlet pressure profiles that these two equations define are shown in Figure 2.

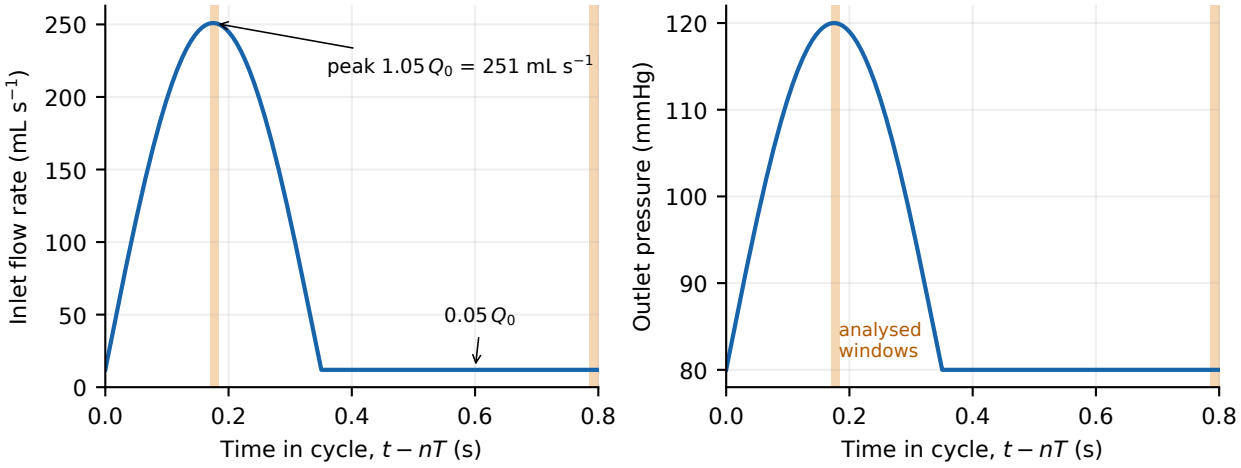


Figure 2: Prescribed inlet flow rate (left) and outlet pressure (right) over one cardiac cycle.

A no-slip condition was imposed at the wall.

2.2.4 Hemodynamic parameters

WSS denotes the tangential force per unit area applied by blood flow against the vessel wall. The WSS for vessels [32] can be calculated using the formula:

$$\tau_w = \mu \left. \frac{\partial u_t}{\partial n} \right|_{n=0}, \tag{2.10}$$

where μ is the blood viscosity, $\frac{\partial u_t}{\partial n}$ is the tangential velocity gradient in the direction n normal to the wall, and u_t is the local tangential velocity component at the wall.

The cycle-averaged magnitude of WSS was computed on the aneurysm wall [50, 51]:

$$\text{TAWSS} = \frac{1}{T} \int_0^T |\mathbf{WSS}(\mathbf{s}, t)| \, dt, \tag{2.11}$$

where T is the duration of one cardiac cycle and $\mathbf{s}$ denotes a point on the vessel wall. TAWSS represents the magnitude of the WSS vector averaged over time.

TKE per unit mass, $k = \frac{1}{2}\overline{v_i' v_i'}$, was used to summarize the modeled velocity fluctuations in the sac. It is a scalar quantity with units of $\mathrm{m}^2\ \mathrm{s}^{-2}$, rather than the fluctuating velocity components themselves.

Speed and TKE were also summarized at eight cross-sectional planes (D1–D8) to compare their variation along the vessel with local diameter.

Maximum principal stress (MPS) was evaluated on the inner wall surface to describe tensile loading in the structural model. It is the largest principal value of the local stress tensor. Elevated MPS identifies modeled stress concentrations; predicting failure would additionally require local tissue-strength data, which were not available.

2.2.5 CFD simulation

ANSYS Fluent, a commercial finite-volume solver with a pressure-based formulation, was used to solve the governing equations for mass and momentum conservation. Blood was modeled as an incompressible Newtonian fluid with density $\rho = 1050\,\mathrm{kg\,m^{-3}}$ and dynamic viscosity $\mu = 3.5 \times 10^{-3}\,\mathrm{Pa\,s}$.

The spatial convergence of the steady flow results was checked by performing a mesh refinement study. Prescribed flow rate was $2.39 \times 10^{-4}\,\mathrm{m^3/s}$ and the SST k-ω turbulence model was used. Four different meshes (2.9 mm to 0.8 mm) were used in the testing as shown in (Table 4). The relative change, Richardson-extrapolated value and grid convergence index (GCI) were computed between each mesh and the immediately finer mesh. The assumed second order convergence rate, $p = 2$ was used in conjunction with Roache's conservative safety factor, $F_s = 3.0$ for the evaluation of the GCI. The two first refinements had relatively high GCI values, which means that the coarser meshes were not in the range of mesh independent solution. For comparison, refinement from 1.3 mm to 0.8 mm changed the maximum aneurysmal WSS by only 0.89%, was Richardson extrapolated to 66.83 Pa, and had a fine-grid GCI value of 1.63%. Therefore, the 0.8 mm mesh was selected for the final simulations.

Table 4: Reported steady-flow fluid-mesh refinement for maximum aneurysmal WSS. The extrapolation assumes second-order convergence.

Element size (mm)	Number of elements	Maximum aneurysm WSS (Pa)	Relative change (%)	Richardson-extrapolated WSS (Pa)	GCI_{fine} (%)
2.9	62 261	47.8	–	–	–
1.9	196 142	55.4	13.72	61.12	30.95
1.3	578 040	67.8	18.29	78.71	48.29
0.8	1 475 430	67.2	0.89	66.83	1.63

The temporal convergence was evaluated for three different time step sizes: 0.01 s, 0.001 s, and 0.0005 s. The difference between the two smaller time steps was small, less than 2%, and there was very good agreement in the WSS waveforms. The time step was chosen as 0.001 s, which was found to be adequate in terms of both the level of temporal resolution and computational efficiency.

To eliminate initial transient effects and ensure a periodic steady state, the simulations were run for four consecutive cardiac cycles ($n = 0, 1, 2, 3$). The percentage difference in WSS between the third and fourth cycles was negligible (mean $< 0.4\%$), confirming that a periodic steady state had been reached. Accordingly, snapshots from the third cardiac cycle were used for the instantaneous flow comparisons. The exact averaging cycle of the retained TAWSS export requires separate confirmation.

2.2.6 Pressure-loaded structural model

The arterial wall was modeled using ANSYS mechanical and considered as a nearly incompressible, isotropic hyperelastic material using the second-order Yeoh strain energy potential. The strain energy density function is

$$W = C_{10}(\bar{I}_1 - 3) + C_{20}(\bar{I}_1 - 3)^2, \tag{2.12}$$

where W is the strain energy density, $\bar{I}_1$ is the first deviatoric strain invariant, and C_{10} and C_{20} are material constants. The material constants were set to $C_{10} = 98.8$ kPa and $C_{20} = 779.1$ kPa [52]. The incompressibility parameter D_1 was specified as 2.04×10^{-4} kPa^{-1}, corresponding to a Poisson's ratio of $\nu = 0.49$. A uniform wall thickness of 2.0 mm was assumed for all models, consistent with previous idealized FSI studies of thoracic aortic aneurysms (AAs), and the arterial wall density was set to 1120 kg/m^3.

Both ends of the structural domain were fixed. Fluid pressure was transferred to the inner wall surface in ANSYS Mechanical (Figure 3). All quantitative flow and reconstruction results use the rigid-wall fluid solution (Section 3.1). The retained coupling records are insufficient to establish a bidirectional moving-wall calculation; the structural results are therefore interpreted as pressure-loaded wall-stress fields.

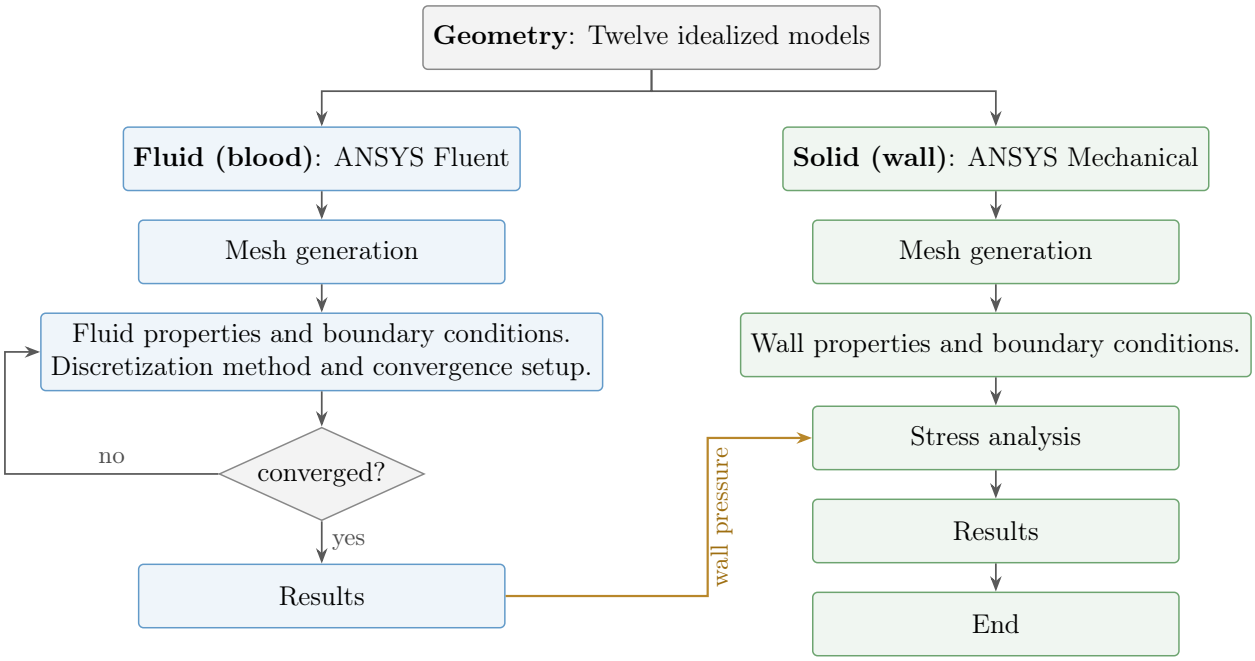


Figure 3: CFD and structural workflow. The fluid problem is solved in ANSYS Fluent, and the wall pressure is transferred to the structural model in ANSYS Mechanical.

3 Sparse-velocity reconstruction methods

3.1 Exported flow fields, instants and regions

All quantitative flow results in this paper, including the reconstruction study, use whole-domain exports of the rigid-wall CFD solutions rather than the structural calculation. For each case the exports contain every fluid node (468 000–603 000 interior nodes and 34 600–41 400 wall nodes) with velocity, static pressure, eddy viscosity μ_t, k and ω, together with the WSS vector on the wall nodes and the inlet and outlet faces. The mesh and node ordering are identical across the snapshots of a case, so differences between snapshots are exact nodal differences. No cell volumes, face areas or connectivity were exported, and all averages are therefore node-weighted. The near-wall node spacing differs between the two groups of models. In the aneurysmal models the interior node nearest each wall node lies a median 0.03–0.04 mm from the wall in the aneurysm zone but 0.58 mm on the rest of the wall, including the bend, whereas in the controls it lies 0.04 mm from the wall everywhere. Near-wall refinement in the aneurysmal models is therefore confined to the aneurysm zone, and quantities outside that zone, including comparisons with the controls there, depend on near-wall resolution as well as on geometry.

Snapshots exist for every case at 1.780 s and at 2.390 and 2.400 s, and for all cases except Case 2 at 1.775 s; Case 2 instead has a snapshot at 1.788 s, which completes its systolic window. Peak inflow occurs at 1.775 s, 0.175 s into the third cycle, so 1.780 s is 5 ms after the peak, and 2.400 s is the end of the diastolic plateau of the third cycle. We use 1.780 s and 2.400 s as common systolic and diastolic instants and refer to them below as the systolic and diastolic phases. All exported fields were post-processed and plotted with Python scripts (NumPy, SciPy and Matplotlib). The earlier case-specific image exports were taken at different near-systolic times (1.775–1.815 s), and they are used only as qualitative

illustrations.

Two regions are used throughout. The *aneurysm zone* is the wall zone exported as the dilation for each aneurysmal case, which spans the sac and its necks. For comparisons that include the controls, the *dilation segment* is the axial extent of the aneurysm zone of the axisymmetric case with the same inlet diameter, restricted to the descending branch, so that each control is evaluated over the corresponding stretch of undilated vessel. Within the straight descending segment the global x axis coincides with the local downstream direction, which makes a negative x velocity a meaningful indicator of reversed flow there, unlike in the curved inlet region.

3.2 Task and information available to each method

Each reconstruction problem concerns one geometry and one short window of two snapshots. The systolic window is 1.775–1.780 s, with the target at 1.780 s, except for Case 2, which has no 1.775 s export and uses 1.780–1.788 s with the same target. The diastolic window is 2.390–2.400 s, with the target at 2.400 s. Given velocity at a sparse set of interior nodes at both window times, a method must recover velocity at all other interior nodes, the gauge pressure in the interior and the WSS magnitude on the aneurysm zone at the target time. The geometry is known, so the task is reconstruction within a given anatomy rather than prediction for a new one.

Table 5 lists what each method may use. Geometry comprises the coordinates of the interior nodes without their field values, the wall nodes with inward normals, and the inlet and outlet faces. Boundary information comprises the prescribed inlet velocity and the mean outlet pressure at each window time. Pressure, WSS and velocity at unobserved nodes are never used for fitting or selection. We examine two viscosity settings. The first supplies the CFD eddy-viscosity field μ_t and its spatial gradient, so the momentum balance uses $\mu + \mu_t$. We call this the *oracle* setting because it gives the reconstruction access to closure information from the reference simulation. The second uses only the prescribed blood viscosity, $\mu = 3.5 \times 10^{-3}$ Pa s, with $\mu_t = 0$; this is the *molecular-viscosity* setting. The CFD eddy viscosity is not measured by the sparse velocity observations. Results that use it are therefore conditional on that additional information.

3.3 Observation design

Velocity is observed at the interior nodes nearest to the points of a uniform Cartesian grid with spacing Δ and a random offset, keeping only grid points with an interior node within $\Delta/2$. The same nodes are observed at both window times. Sampling on a grid rather than on mesh nodes avoids oversampling the refined near-wall layers. Each grid point is, however, snapped to a mesh node, so observations next to the wall fall on the prism layers. At $\Delta = 2.5$ mm, 5.6–8.5% of the observed nodes of Cases 2–8 lie within 0.25 mm of the wall and 26–30% within 1 mm, the closest at 0.03 mm, whereas a 2.5 mm imaging voxel would return an average over its volume. The design is therefore point sampling on a grid rather than voxel sampling. Every method receives the same observations, so the comparison between methods is not affected, but the absolute WSS errors are optimistic relative to imaging data; Section 5.7 quantifies this by removing the near-wall observations. The primary spacing

Table 5: Information used by each method. All methods receive the same observation nodes at both window times and are scored through the same post-processing.

Information	Linear	RBF	Data field	Continuity field	Momentum PINN
Sparse velocity at two times	✓	✓	✓	✓	✓
Wall no-slip and inlet velocity	✓	✓	✓	✓	✓
Outlet pressure datum	post-proc.	post-proc.	✓	✓	✓
Interior node coordinates (no values)	–	–	–	✓	✓
Continuity in the fit	–	–	–	✓	✓
Momentum balance in the fit	–	–	–	–	✓
CFD μ_t (oracle setting)	post-proc.	post-proc.	post-proc.	post-proc.	fit + post-proc.

$\Delta = 2.5$ mm observes 1.27–1.36% of the interior nodes per time (5 970–8 205 nodes), and the secondary spacings of 1.5 and 4.0 mm observe 5.4–5.8% and 0.34–0.37%. The observed values are noise-free CFD values at mesh nodes. The neural fields withhold a random 10% of sampled nodes from fitting for model selection and monitoring. The fixed interpolation comparators use all sampled observations; the tuned comparators use the holdout for selection and then refit using all observations. Thus the sampled observation pool is shared, but the final fitting subsets differ. Each draw s fixes both the grid offset and the network initialization, draws are paired across methods, and the observed node indices are stored.

3.4 Space-time neural fields and physics constraints

We use a space-time neural field to represent each short flow window. The confirmation method is a continuity-constrained field followed by physics-based post-processing, in which the network reconstructs velocity and separate calculations recover pressure and WSS. We refer to this method as the continuity field. Its divergence penalty is a physical constraint, but it does not enforce the momentum balance during fitting. We use the term *momentum PINN* for the separate arms, tested only on the two development geometries, that also penalize the momentum residual.

All neural arms share the same architecture, which maps position and time to velocity and pressure, $(\mathbf{x}, t) \mapsto (\mathbf{u}_\theta, p_\theta)$. Coordinates are scaled as $\mathbf{x}_s = (\mathbf{x} - \mathbf{x}_c)/L$, where $\mathbf{x}_c$ is the centroid of the interior nodes and L is half the largest extent of the domain, and time is mapped to $t_s \in [-1, 1]$ across the window. Velocity is scaled by U, the 99.5th percentile of training-observation speed at the target time, and pressure by ρU^2. The input is $[\sin(2\pi \mathbf{B}^\mathsf{T}\mathbf{x}_s), \cos(2\pi \mathbf{B}^\mathsf{T}\mathbf{x}_s), \mathbf{x}_s, t_s]$, where $\mathbf{B} \in \mathbb{R}^{3\times 64}$ holds Gaussian Fourier features with a standard deviation (SD) of 2 [53], followed by six fully connected layers of width 256 with SiLU activations and a linear output of four scaled quantities (about 360 000 parameters) (Figure 4).

The loss combines data and boundary terms with an optional physics term,

$$\mathcal{L} = \mathcal{L}_{\text{obs}} + \mathcal{L}_{\text{wall}} + \mathcal{L}_{\text{in}} + \mathcal{L}_{\text{out}} + w\,\mathcal{L}_{\text{phys}}, \tag{3.1}$$

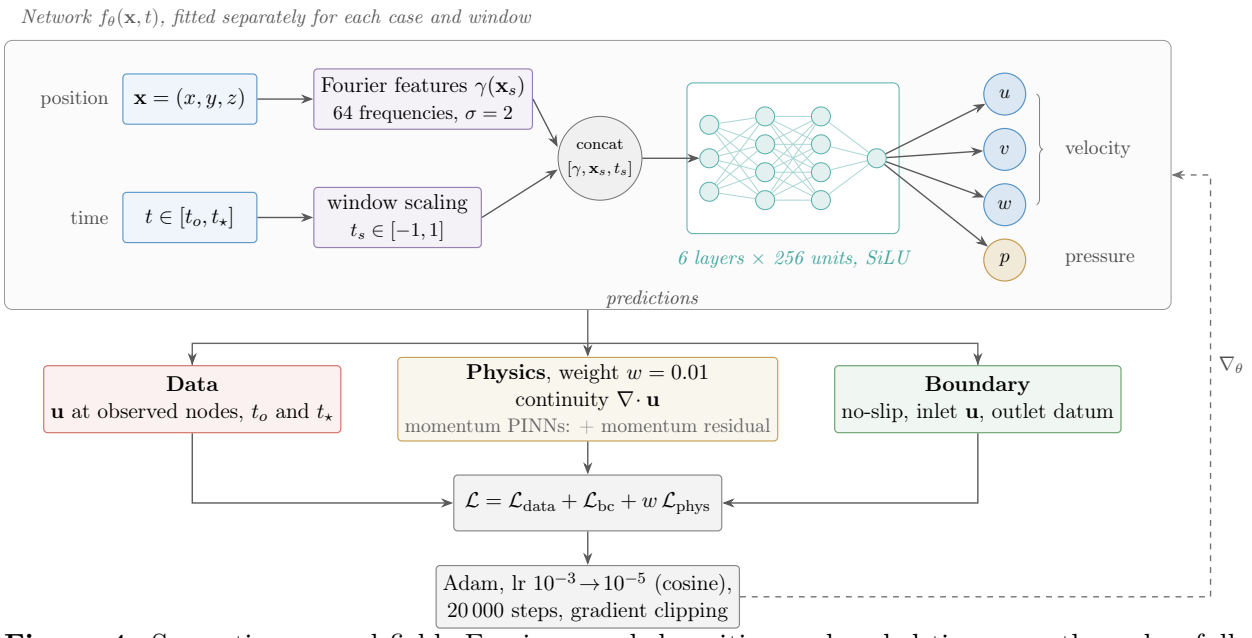


Figure 4: Space-time neural field. Fourier-encoded position and scaled time pass through a fully connected network to velocity and pressure, fitted to the observed velocity and the boundary conditions. The continuity arm adds a divergence penalty; the momentum residual (gray) was used only by the momentum PINNs tested on the two development geometries.

where $\mathcal{L}_{\text{obs}}$ is the mean squared velocity misfit at the observed nodes at both times, $\mathcal{L}_{\text{wall}}$ penalizes velocity at wall nodes at random times in the window, $\mathcal{L}_{\text{in}}$ penalizes the misfit to the prescribed inlet velocity at each window time, and $\mathcal{L}_{\text{out}}$ penalizes the squared mean of p_θ on the outlet face, which fixes the pressure datum. Four arms differ only in $\mathcal{L}_{\text{phys}}$. The *data* arm has none. The *continuity* arm uses the mean squared divergence $(\nabla \cdot \mathbf{u}_\theta)^2$ at interior nodes drawn at random times. The *steady* and *unsteady* momentum arms add the mean squared residual of

$$\rho\,\partial_t \mathbf{u} + \rho(\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p - \nabla \cdot \left[(\mu + \mu_t)(\nabla \mathbf{u} + \nabla \mathbf{u}^{\mathsf{T}})\right] = \mathbf{0}, \tag{3.2}$$

without the time term in the steady arm, with μ_t interpolated linearly in time between the two snapshots in the oracle setting and set to zero in the molecular setting. Derivatives are taken by automatic differentiation, and the variable-viscosity divergence retains the gradient of μ_t. Equation (3.2) has no separate $2\rho k/3$ term because the exported CFD pressure satisfies the momentum balance better without it (Section 5.1), which is consistent with an effective-pressure convention but does not, by itself, verify the solver's exported pressure definition.

Each step draws 4096 observations, 4096 collocation nodes, 2048 wall nodes, 512 inlet nodes per time and 256 outlet nodes per time. Training uses Adam with a learning rate of 10^{-3}, cosine decay to 10^{-5} over 20 000 steps and gradient-norm clipping at 1. There is no early stopping, and the last iterate is scored. The physics weight w was chosen for each arm from $\{0.01, 0.1, 1\}$ on Case 1 at systole by the error on the withheld observations and then frozen. That error increased with w in every arm, so $w = 0.01$ was used throughout. On one NVIDIA Quadro RTX 8000 shared with other jobs, a 20 000-step fit took a median of

23 min for the continuity arm and 17 min for the data arm.

3.5 Interpolation comparators

The comparators interpolate the same observations, with zero velocity at the wall nodes and the prescribed inlet velocity added as sources, separately at each window time. *Linear* interpolation uses the Delaunay triangulation of the sources, with the nearest source used outside its hull. *RBF* interpolation uses a local thin-plate-spline RBF with a linear polynomial on the 64 nearest sources, with fallback neighborhoods of 128 and then 256 sources if the local system is singular. These settings were fixed before the development test and are the prespecified comparator. Two further comparators were added after the confirmatory verdict as robustness checks. The *tuned RBF* selects the kernel (thin-plate spline, cubic or quintic), the neighborhood size (32–256) and the smoothing parameter (0–10^{-2}) by the error on the withheld observations. The *space-time RBF* fits one interpolant over $(\mathbf{x}, c\,t)$ through both snapshots, with the time scale $c \in \{0, 0.5, 2, 8\}$ m s^{-1}, the kernel and the neighborhood size chosen on the same withheld observations.

3.6 Shared post-processing

Every reconstruction, including the complete CFD velocity used as a reference, is converted to pressure and WSS by the same steps (Figure 5). The reconstructed velocity is first sampled at the CFD nodes. Spatial gradients are computed by weighted least squares on the 40 nearest nodes with inverse-square-distance weights, with zero velocity imposed at the wall nodes. The momentum balance then gives a target pressure gradient,

$$\mathbf{f} = -\rho \frac{\mathbf{u}(t_\star) - \mathbf{u}(t_o)}{t_\star - t_o} - \rho(\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla\cdot\left[(\mu+\mu_t)(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathsf{T}})\right], \tag{3.3}$$

evaluated at the target time $t_\star$, where t_o is the other window time. The gauge pressure is the least-squares solution of $\nabla_h p = \mathbf{f}$ on the interior nodes more than 1 mm from the wall, with the mean pressure over nodes beyond $x = 0.130$ m set to zero, computed with LSQR. Near-wall nodes are excluded because the prism-layer stencils are poorly conditioned. Dropping the first term of Eq. (3.3) gives the steady form, and setting $\mu_t = 0$ gives the molecular form. All four combinations are reported, and the unsteady form with oracle μ_t is the prespecified pressure endpoint. The steady form of Eq. (3.3) is the standard pressure-gradient route of velocity-based relative-pressure estimation [29].

WSS is estimated at each aneurysm-zone wall node $\mathbf{x}_w$ with inward unit normal $\mathbf{n}$ from the tangential velocity $\mathbf{u}_t$ at two distances inside the lumen,

$$\boldsymbol{\tau}_w \approx \mu \frac{4\,\mathbf{u}_t(\mathbf{x}_w + h\mathbf{n}) - \mathbf{u}_t(\mathbf{x}_w + 2h\mathbf{n})}{2h}, \tag{3.4}$$

which is second-order accurate for a no-slip wall, with $\mathbf{u}_t$ obtained by linear interpolation of the nodal velocity. The prespecified distance is $h = 0.25$ mm, and results at 0.1 and 0.5 mm are reported only as an oracle sensitivity. In the aneurysm zone the interior node nearest the wall lies a median 0.03–0.04 mm from it, so Eq. (3.4) is informative there. Outside the

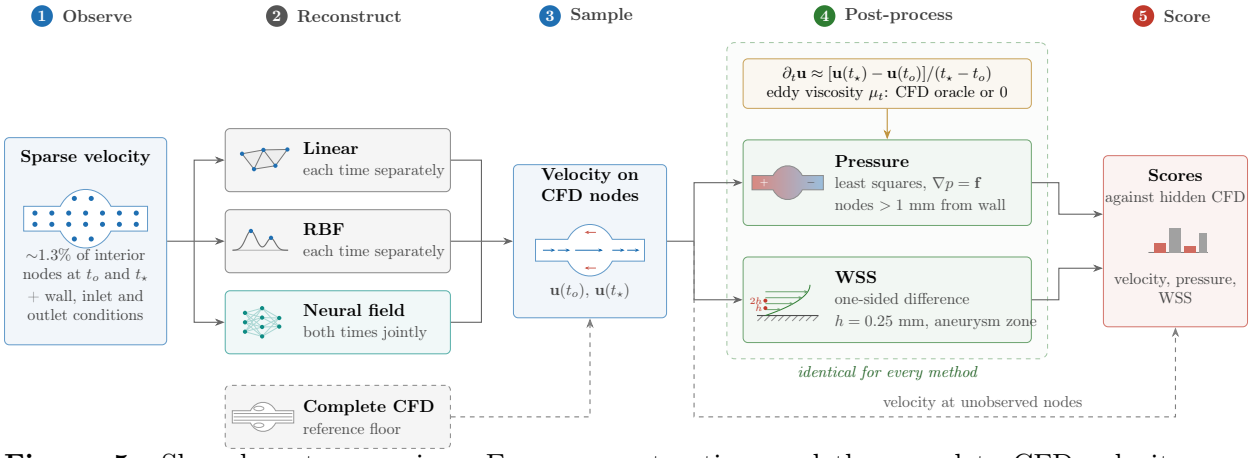


Figure 5: Shared post-processing. Every reconstruction, and the complete CFD velocity as a reference, is converted to pressure and WSS by the same steps, so differences between methods come only from the reconstructed velocity.

aneurysm zone that distance is 0.58 mm, and even the complete CFD velocity reproduces the exported WSS there only to a relative error of 0.46–0.68, so WSS outside the aneurysm zone is not used as a target.

3.7 Error measures

For a field q on an evaluation set we report the node-weighted relative error $\|q_{\text{pred}} - q_{\text{CFD}}\|_2 / \|q_{\text{CFD}}\|_2$. Velocity is scored as a vector on all unobserved interior nodes, pressure on the integration sub-cloud after the common gauge, and WSS as a magnitude on the aneurysm-zone wall nodes. A global relative error does not bound local errors, so we also report the 95th percentile of the pointwise velocity-error magnitude in m s^{-1}, velocity error inside the sac, the ratio of predicted to CFD 99th-percentile WSS, the WSS vector error, and the pressure drop between the inlet and outlet ends of the sub-cloud in Pa. The last matters at diastole, where drops of 0.4–8.8 Pa make relative errors hard to interpret. The *floor* of a quantity is the error obtained when the complete CFD velocity is passed through the same post-processing. It measures the error of the post-processing rather than a strict lower bound, and a reconstruction can fall slightly below it.

3.8 Study protocol

The study had three stages, and each was specified in writing before its runs. First, the complete CFD fields of all nine aneurysmal cases were used to check which form of Eq. (3.2) the exported fields satisfy and to measure the post-processing floors. This stage used the complete fields of Cases 2–8, pressure and WSS included, to fix the shared post-processing, namely the momentum form and the WSS target region. The same choices are applied to every method. However, because the evaluation-case targets informed these choices, the seven-case evaluation is independent of reconstruction tuning rather than of the entire analysis pipeline. Second, a development test on two geometries kept out of the confirmation set, Case 1 (2.0 cm, axisymmetric) and Case 9 (2.6 cm, posterior-dominant), compared the four neural arms, the unsteady arm with molecular viscosity, and linear and RBF interpolation,

with two paired draws at systole and one at diastole. Its gate required the unsteady PINN with oracle μ_t to reduce both the pressure and the WSS error by at least 15% relative to the best non-PINN method with the same information, in both cases at systole, with velocity no worse than 1.1 times the best interpolation.

The development test failed (Section 5.2), and the confirmation stage tested the best-performing staged method that emerged from it on the seven aneurysmal cases not used to fit, select or tune any reconstruction method, Cases 2–8. The candidate is the continuity field followed by the shared post-processing, with nothing retuned from the development test. The primary comparison is against RBF followed by the same post-processing, at $\Delta = 2.5$ mm with three paired draws, over 14 case–phase units. Each unit's error is the mean over its three draws. The two hypotheses, fixed before any confirmation run, were that the candidate has the lower pressure error (unsteady form with oracle μ_t) in at least 12 of 14 units with a median error ratio to RBF of at most 0.6, and the lower aneurysm-zone WSS error at $h = 0.25$ mm in at least 12 of 14 units with a median ratio of at most 0.8. A guardrail required hidden-velocity error no larger than RBF's in every unit. The data arm, linear interpolation, the four pressure forms, the 1.5 and 4.0 mm budgets (one draw each) and the floors are secondary analyses, reported regardless of their outcome. The tuned and space-time RBF were added after the verdict and are reported as robustness checks outside the confirmatory claim.

4 Results and discussion

The geometry comparisons use nine aneurysmal models and three matched controls, grouped by inlet diameter (Table 2). We report streamlines, pressure drop, reversed-flow fraction, WSS, TAWSS and structural MPS. Instantaneous flow quantities use the common systolic and diastolic times defined in Section 3.1. The structural fields represent systolic pressure loading; TAWSS is a cycle average, whose exact averaging cycle requires confirmation from the solver record.

4.1 Flow structure

Figures 6–8 show recirculation in the aneurysm sacs and at the inner curvature of the proximal descending aorta. At the bend, the flow shifts toward the outer curvature, leaving a low-momentum region along the inner wall where separation occurs; we refer to this as the inner curvature recirculation zone (ICRZ) (Figure 1). Within the sacs, the recirculation pattern changes with bulge direction and is pronounced in the anterior-dominant geometries. The transverse views show secondary vortices on both sides of the sac. The controls have smoother streamlines apart from the bend-associated recirculation. These patterns are consistent with prolonged residence in parts of the sac, but thrombus formation and tissue weakening were not simulated [54].

4.2 Pressure drop, reversed flow and eddy viscosity

Table 6 summarizes the common-time flow descriptors for all twelve geometries, and inlet diameter clearly dominates the systolic response. The inlet-to-outlet pressure drop in the

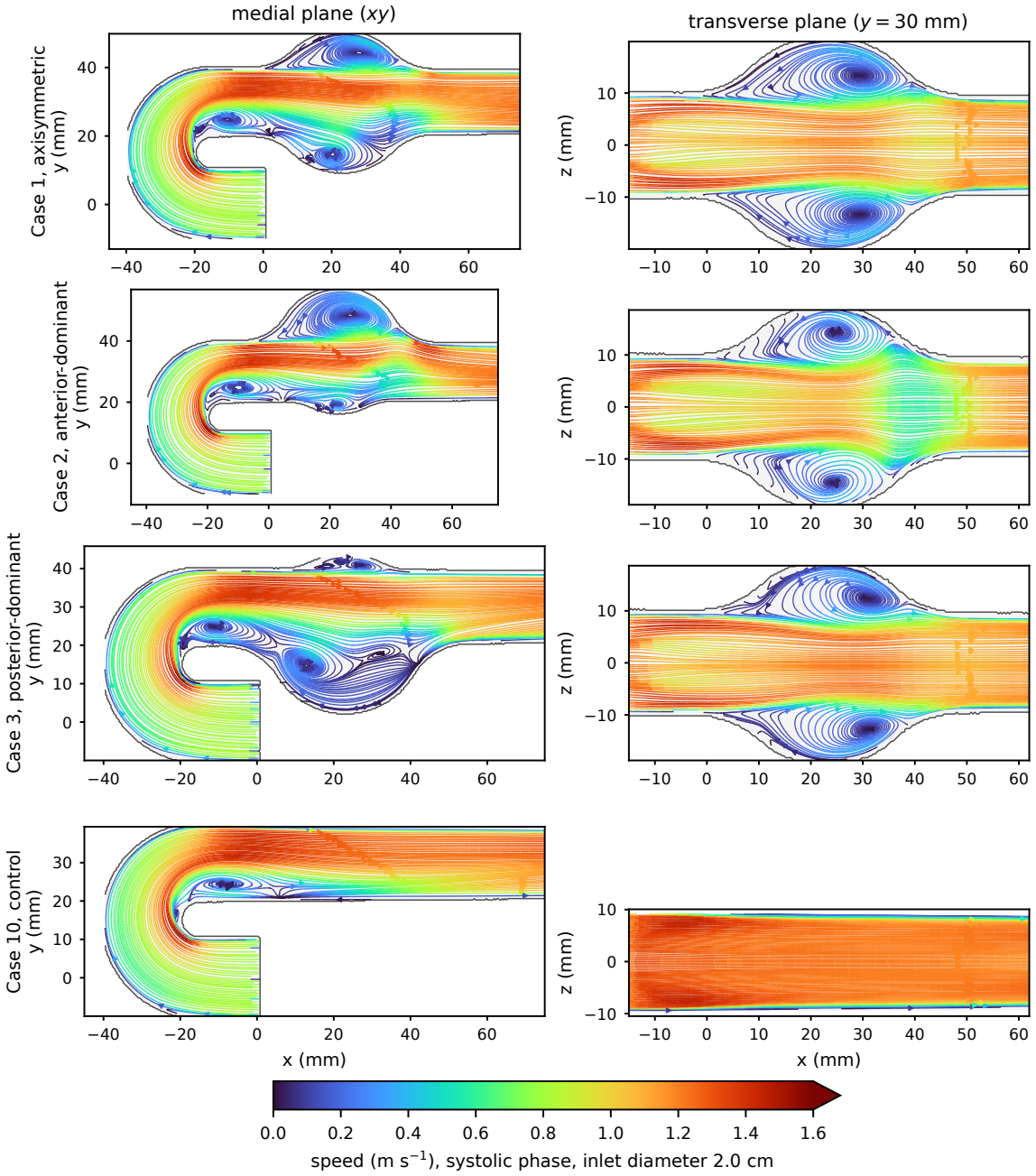


Figure 6: Streamlines in the systolic phase for the 2.0 cm geometries in the medial plane (left) and the transverse plane (right), colored by in-plane speed on one scale for all geometries. Rows are the axisymmetric, anterior-dominant and posterior-dominant sacs and the control.

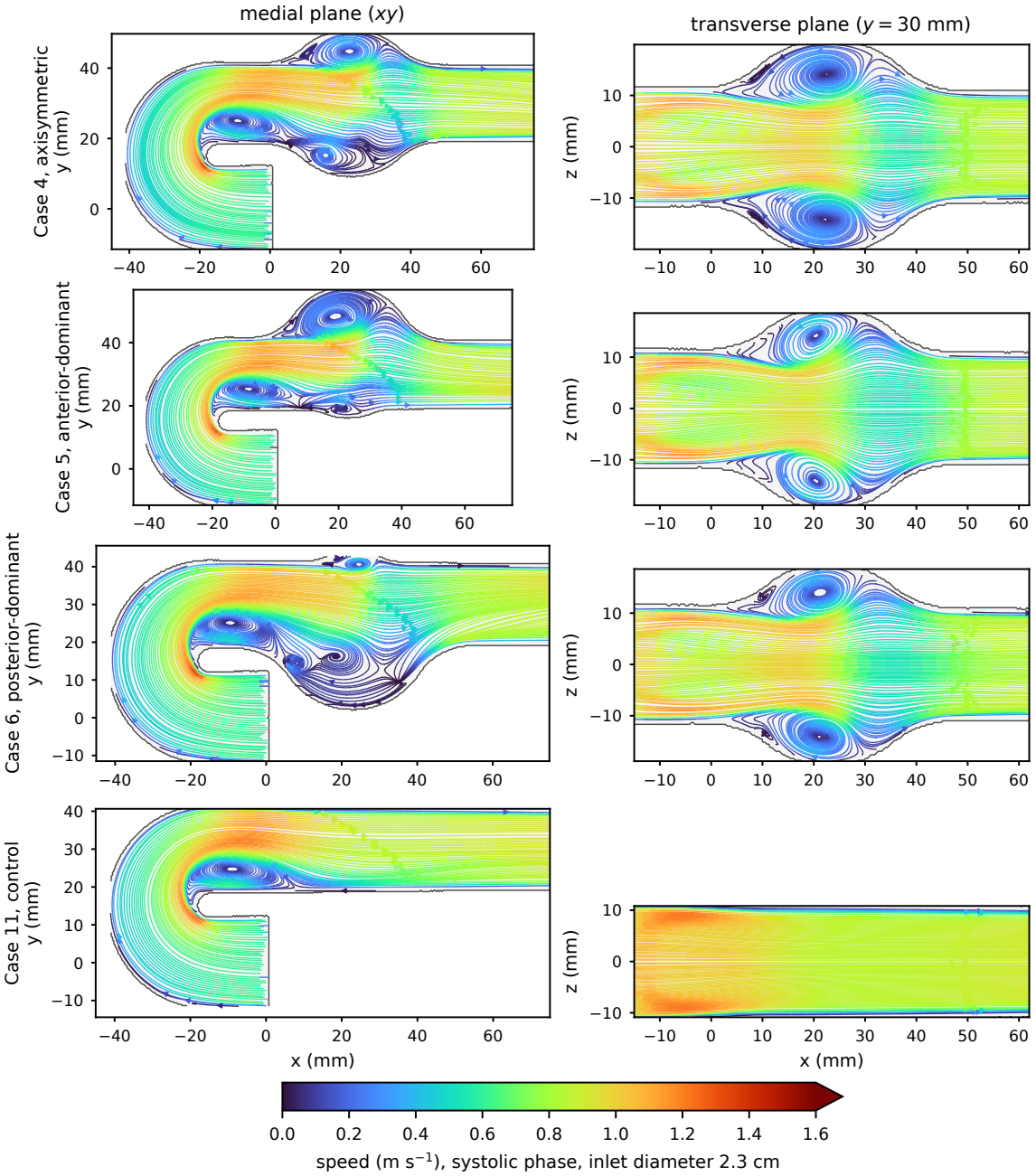


Figure 7: As Figure 6, for the 2.3 cm geometries.

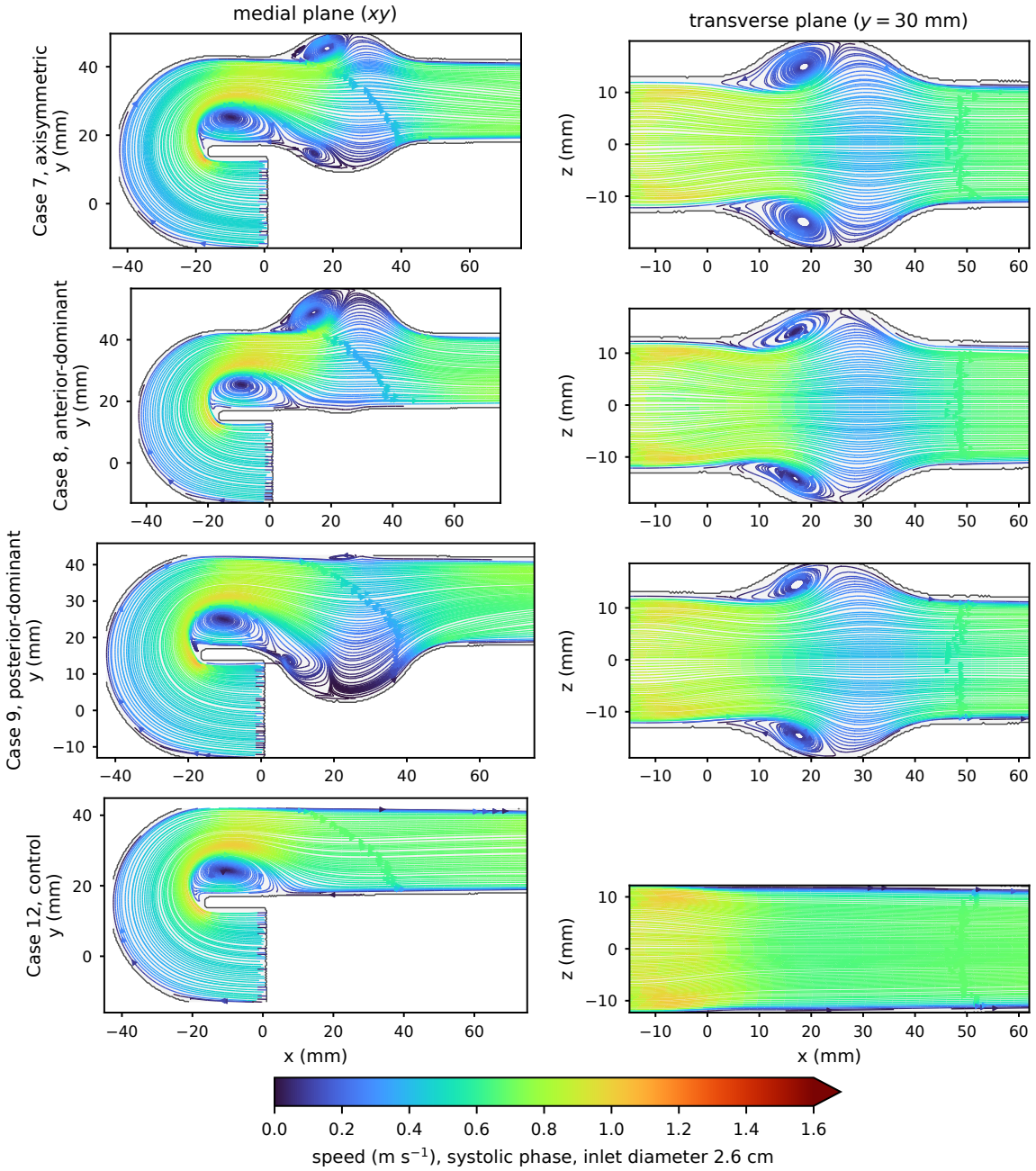


Figure 8: As Figure 6, for the 2.6 cm geometries.

systolic phase decreases from 1261–1346 Pa at 2.0 cm to 844–870 Pa at 2.3 cm and 528–561 Pa at 2.6 cm, whereas at a fixed diameter the three bulge patterns differ by at most 7%. Relative to the matched control, the dilation adds 17–30% to the pressure drop. The fraction of dilation-segment nodes exhibiting reversed axial velocity at systole follows the same ordering, falling from 50–54% at 2.0 cm to 38–40% at 2.3 cm and 24–28% at 2.6 cm, against only 2–3% in the controls. The mean eddy viscosity within the segment is 8.6–9.5 times the molecular viscosity at 2.0 cm but only 1.6–2.1 times at 2.6 cm. A narrower parent vessel therefore delivers a faster jet into the same 4.0 cm sac, producing a larger expansion, a greater fraction of reversed flow, and stronger modeled turbulent mixing. At diastole, by contrast, the pressure drops are only 1.6–10.7 Pa, and 35–47% of the segment carries reversed flow in every geometry, including the controls, because the slow plateau flow no longer possesses a coherent core.

Table 6: Common-time descriptors of the rigid-wall CFD at systole (1.780 s) and diastole (2.400 s). Δp is the difference between the mean static pressure on the inlet and outlet faces, and the reversed-flow fraction is the fraction of nodes with $u_x < 0$, the local downstream direction in the straight segment. The evaluation segment is the dilation segment of Section 3.1; all values are node-weighted descriptive summaries of single designed geometries.

			Δp (Pa)		Systolic τ_w (Pa)		Diastolic τ_w (Pa)		Mean speed	Reversed flow (%)		$\overline{\mu_t}/\mu$
Case	D_{in} (cm)	β	Sys.	Dia.	Median	95th pct.	Median	95th pct.	sys. (m s^{-1})	Sys.	Dia.	sys.
1	2.0	1.00	1261	10.7	1.94	17.6	0.279	0.60	0.362	50	35	9.5
2	2.0	0.48	1346	3.1	2.83	17.4	0.729	1.07	0.391	50	45	9.4
3	2.0	2.08	1297	5.7	1.65	15.4	0.851	1.24	0.346	54	45	8.6
4	2.3	1.00	844	6.8	2.30	11.7	0.253	0.46	0.328	39	37	3.6
5	2.3	0.48	870	1.7	2.39	10.7	0.613	0.98	0.330	38	46	4.0
6	2.3	2.08	870	7.0	2.06	10.8	0.486	0.89	0.316	40	43	3.2
7	2.6	1.00	545	4.9	1.87	8.1	0.251	0.59	0.275	28	38	1.6
8	2.6	0.48	561	1.6	1.85	7.4	0.515	0.83	0.274	28	46	2.0
9	2.6	2.08	528	5.8	1.82	7.1	0.447	0.91	0.261	24	47	2.1
10	2.0	Control	1074	6.6	5.02	8.8	0.125	0.52	0.730	2	35	4.2
11	2.3	Control	668	2.8	3.53	4.9	0.103	0.38	0.546	3	41	2.6
12	2.6	Control	448	1.7	2.65	3.5	0.172	0.30	0.434	3	43	2.0

4.3 Wall shear stress

Figure 9 illustrates the WSS magnitude over the dilation segment in the systolic phase, unwrapped about the axis of the descending segment. In this representation the anterior wall lies at 0° and the posterior wall at ±180°, so that the full circumference of the vessel is displayed on a single plane. At systole all aneurysmal cases share a low-shear sac body bounded by high-shear rings at the proximal and distal necks, where the jet enters the sac and reattaches. The median segment WSS is 1.6–2.8 Pa in the aneurysmal cases against 2.7–5.0 Pa in the controls, whereas the 95th percentile is 1.8–2.4 times that of the matched control (Table 6). The neck rings weaken with increasing inlet diameter, and the 95th percentile falls from 15–18 Pa at 2.0 cm to 7–8 Pa at 2.6 cm. Asymmetry moves the pattern rather than scaling it. In the anterior-dominant cases the low-shear region is centered on the anterior wall (0°), whereas in the posterior-dominant cases it is centered on the posterior wall (±180°), so that over the sac body the wall opposite the dominant bulge carries the

higher shear. In Case 2 the median sac-body WSS is 4.5 Pa on the posterior wall against 0.9 Pa on the anterior bulge, and in Case 3 the 95th percentile is 8.4 Pa on the anterior wall against 2.1 Pa on the posterior bulge, with the same ordering in Cases 5, 6 and 9 and nearly equal values in Case 8. This behavior is consistent with the streamlines, which show that the jet crossing the sac impinges on the wall opposite the bulge while the bulge itself is filled by slow recirculating flow. At the distal neck the highest shear lies on the anterior (outer-curvature) side in most cases, irrespective of the bulge direction, because the core leaves the bend along the anterior wall.

Figure 10 presents the same WSS magnitude in the diastolic phase on one color scale per phase. At diastole the effect of asymmetry becomes larger than that of diameter. The median segment WSS of the asymmetric sacs is 0.45–0.85 Pa, which is 1.8–3.1 times that of the axisymmetric sacs (0.25–0.28 Pa) at every diameter and 2.6–6.8 times that of the controls (0.10–0.17 Pa). On comparison of all cases, the diastolic maps show broad shear over the whole wall of the asymmetric sacs, consistent with a residual vortex that keeps moving along the sac wall during the plateau, whereas the axisymmetric sacs and the controls are close to stagnant. These are instantaneous values at two instants; the cycle-averaged picture is given by the TAWSS maps in Section 4.4.

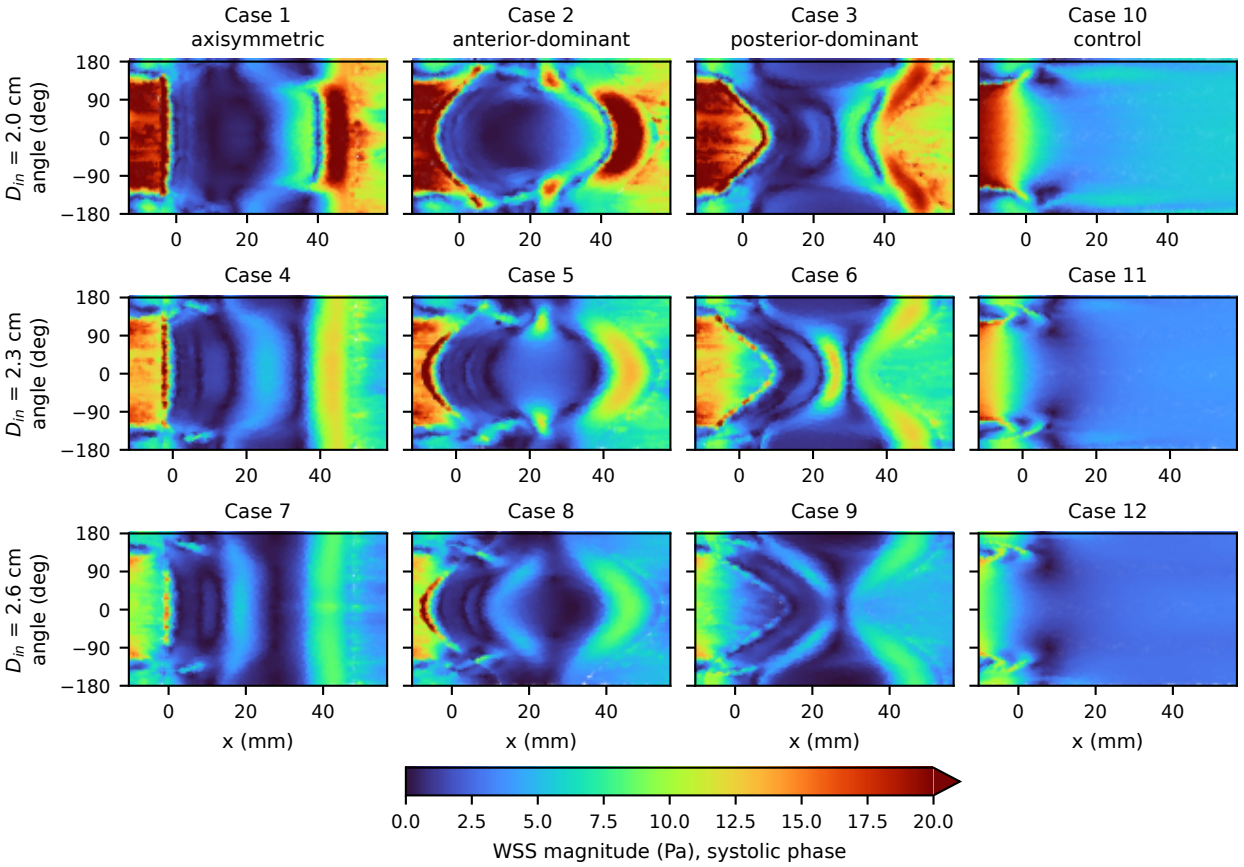


Figure 9: WSS magnitude on the dilation segment in the systolic phase, on one color scale saturating at 20 Pa. The wall is unwrapped about the vessel axis, with the anterior wall at 0° and the posterior wall at ±180°. Rows are inlet diameters and the last column is the matching control. White points are gaps in the exported wall nodes.

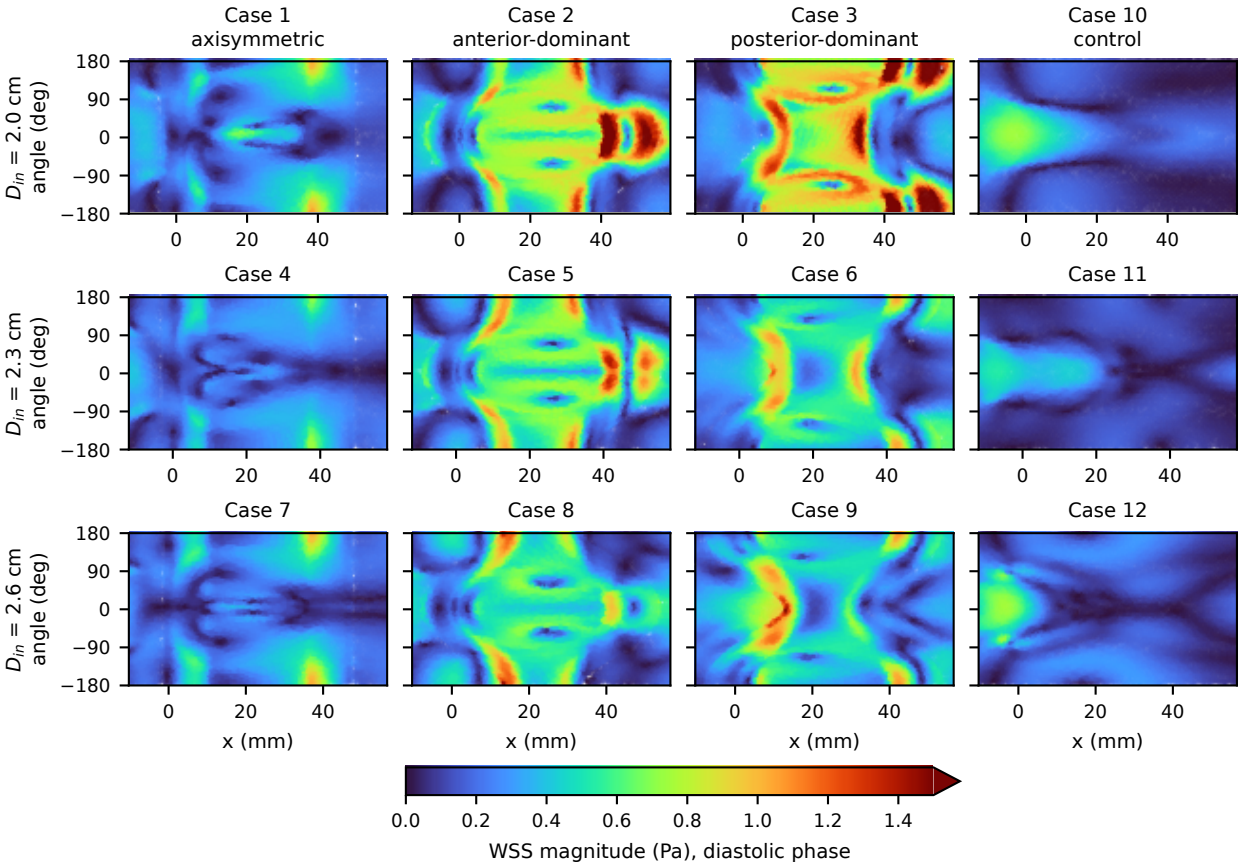


Figure 10: As Figure 9, in the diastolic phase, with one color scale saturating at 1.5 Pa.

Figure 11 shows the signed axial component of WSS for both the medial and transverse planes. Panel (a) shows the WSS on the anterior wall of the medial plane while panel (b) shows the WSS on the posterior wall of the medial plane. Panel (c) shows the WSS for the transverse plane. Because the aorta bends, a negative value alone does not prove local reverse flow in the curved segment, but along the straight descending segment the negative regions coincide with the recirculation seen in Figures 6–8. On the anterior medial wall and on the transverse wall, WSS is large and positive at the proximal neck in every case. It becomes negative where the flow detaches on entering the sac, between $x \approx -4$ and 16 mm on the anterior wall and between 0 and 4 mm on the transverse wall, and it stays negative over the recirculating part of the sac. It returns to positive values, marking reattachment of the core flow, between $x \approx 27$ and 42 mm on the anterior wall and between 26 and 39 mm on the transverse wall, followed by the high positive values of the distal neck. The reattachment point moves upstream as the inlet diameter increases, from 34–42 mm at 2.0 cm to 27–31 mm at 2.6 cm on the anterior wall and from 36–39 mm to 26–27 mm on the transverse wall, so the recirculation region shortens with a wider parent vessel, in line with the reversed-flow fractions of Table 6. The posterior medial wall differs because WSS is already negative at the proximal neck in every geometry, the controls included, which identifies this region as the ICRZ rather than as part of the sac recirculation. The depth of the negative region does not follow the bulge pattern consistently; the most negative value inside the sac, −11.3 Pa, occurs on the anterior wall of Case 6.

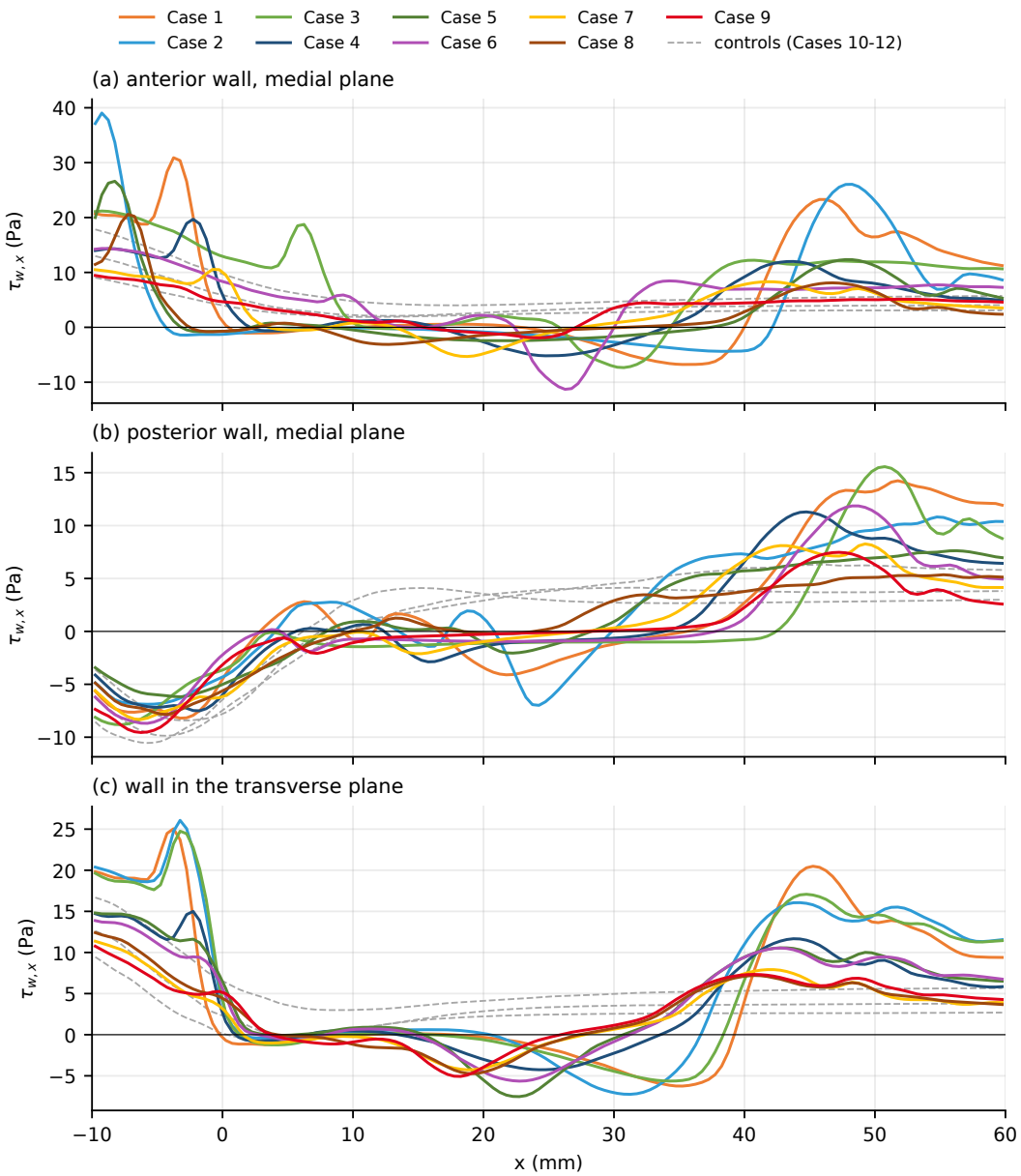


Figure 11: Axial component of WSS (Pa) along wall intersections with the medial and transverse planes at systole: (a) anterior medial wall, (b) posterior medial wall and (c) transverse-plane wall intersection.

4.4 Time-averaged wall shear stress

Figure 12 and Table 7 summarize TAWSS across the twelve geometries. The TAWSS is shown on the dilation segment of every geometry, unwrapped as in Figure 9 and drawn on one logarithmic color scale because the values span two orders of magnitude. Every aneurysmal sac carries a broad region of low TAWSS. The median over the dilation segment is 0.51–0.83 Pa in the aneurysmal cases against 2.5–4.5 Pa over the same stretch of the controls, while over the sac body the median on either side is 0.23–0.84 Pa. Relative to the matched control, the segment median is lower by a factor of 8.5–8.9 at 2.0 cm, 4.0–4.9 at 2.3 cm and 3.4–3.6 at 2.6 cm, so that the reduction in cycle-averaged shear caused by

the dilation is largest for the narrowest parent vessel. The aneurysmal segment medians generally rise from the 2.0 cm group to the larger-diameter groups; the 95th percentiles and control medians decrease with inlet diameter. These statistics therefore describe different trends. The high-shear rings at the necks persist in the cycle average, and their strength falls with inlet diameter, the 95th percentile decreasing from 7.6–10.9 Pa at 2.0 cm to 5.5–6.4 Pa at 2.3 cm and 4.1–5.1 Pa at 2.6 cm.

Aneurysm asymmetry had a strong influence on the spatial distribution of TAWSS. In asymmetric models, the lowest TAWSS is usually located on the dominant bulge. In the anterior-dominant Cases 2, 5 and 8 the sac-body median is 0.34, 0.23 and 0.33 Pa on the anterior wall against 0.52, 0.58 and 0.84 Pa on the posterior wall, whereas in the posterior-dominant Cases 6 and 9 it is 0.24 and 0.28 Pa on the posterior wall against 0.71 and 0.60 Pa on the anterior wall. Case 3 is the exception, with similar values on both sides (0.35 and 0.42 Pa). In the axisymmetric sacs the low-shear region forms circumferential bands with nearly equal anterior and posterior medians. The bulge also has low instantaneous shear at systole. Together, these results show low systolic and low cycle-averaged shear; they do not establish that shear remains low at every instant or identify rupture-prone tissue. Low TAWSS, in particular, marks regions of prolonged low shear, which are known to promote thrombus formation [54], although the present models do not include those processes.

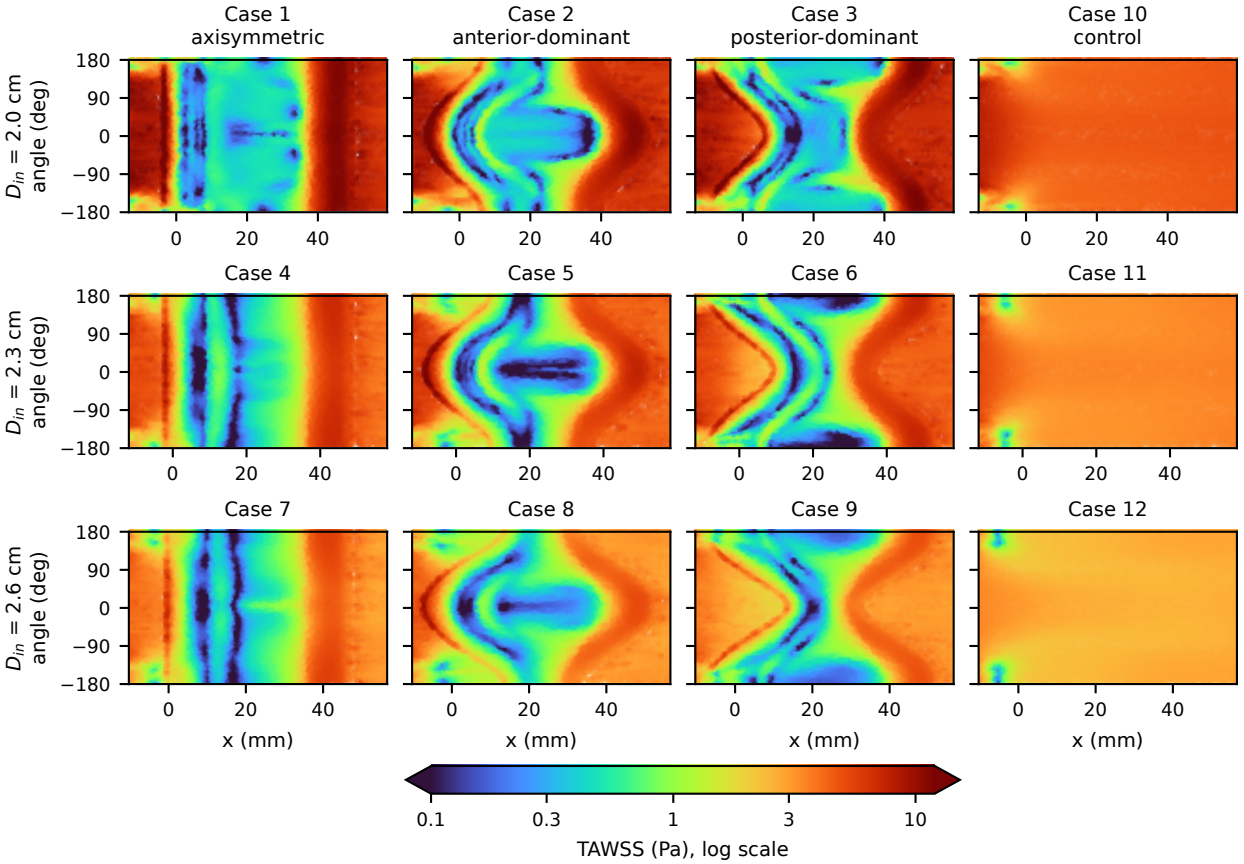


Figure 12: TAWSS on the dilation segment, unwrapped as in Figure 9, on one logarithmic color scale. Rows are inlet diameters and the last column is the matching control.

Table 7: TAWSS (Pa) on the dilation segment. Percentiles are over the whole segment; the sac-body medians use its middle third, within 60° of the anterior or posterior direction.

			Segment			Sac body, median	
Case	D_{in} (cm)	Morphology	5th pct.	Median	95th pct.	Anterior	Posterior
1	2.0	axisymmetric	0.19	0.51	10.86	0.41	0.47
2	2.0	anterior-dominant	0.15	0.52	7.60	0.34	0.52
3	2.0	posterior-dominant	0.18	0.53	8.66	0.35	0.42
10	2.0	control	3.65	4.53	4.89	4.66	4.36
4	2.3	axisymmetric	0.12	0.70	6.41	0.47	0.49
5	2.3	anterior-dominant	0.10	0.67	5.45	0.23	0.58
6	2.3	posterior-dominant	0.10	0.83	5.82	0.71	0.24
11	2.3	control	2.81	3.30	3.52	3.45	3.20
7	2.6	axisymmetric	0.12	0.70	5.06	0.45	0.50
8	2.6	anterior-dominant	0.17	0.73	4.12	0.33	0.84
9	2.6	posterior-dominant	0.18	0.69	4.24	0.60	0.28
12	2.6	control	2.20	2.51	2.82	2.59	2.51

4.5 Structural wall stress

The structural calculation tests how pressure loading is distributed across the prescribed wall geometry. Diameter and asymmetry can alter tissue stress [2], but stress alone does not determine failure because local wall strength is unknown. A Laplace-law estimate based on a uniform cylinder does not capture the changing curvature at an aneurysm's necks and crown; the three-dimensional structural model is used here to describe those stress concentrations.

Figure 13 presents the MPS on the inner surface of the wall over the dilation segment, unwrapped as in Figure 9, and Table 8 summarizes it. In every aneurysmal case the MPS is highest in circumferential bands at the necks, where the wall curvature changes sign, with 95th percentiles of 101–113 kPa over the necks, and lowest over the crown of the sac, where the medians are 55–80 kPa. This pattern is consistent with the findings of Vorp et al. [2]. The two morphology groups differ in their sac-body stress distribution. In the axisymmetric models, the stress was elevated at the proximal regions but lower at the midsection of the aneurysm sac, with similar behavior on both anterior and posterior sides; the anterior and posterior sac-body medians are within 1.2 kPa of each other. In contrast, the asymmetric models exhibited higher stress on the side opposite to the dominant bulge. The sac-body median on the posterior wall of the anterior-dominant Cases 2, 5 and 8 is 69.2, 73.6 and 78.1 kPa against 55.9, 55.0 and 61.8 kPa on the anterior bulge, and the anterior wall of the posterior-dominant Cases 3, 6 and 9 carries 68.3, 72.9 and 80.3 kPa against 55.8, 55.3 and 57.2 kPa on the posterior bulge. Median sac-body MPS on the wall opposite the dominant bulge increases with inlet diameter. The controls show the same trend in segment-median MPS (52.5, 58.4 and 65.1 kPa for inlet diameters of 2.0, 2.3 and 2.6 cm, respectively). This trend is consistent with greater tensile loading at a larger vessel radius under comparable pressure and wall thickness.

The controls have relatively uniform wall stress, with 95th percentiles within 3 kPa of their medians and mild elevations near the bend. In the asymmetric sacs, the wall opposite the dominant bulge has the higher sac-body median. Case 9 has the highest anterior sac-

body median, 80.3 kPa; this is not a claim about the largest local peak. Indeed, the highest reported neck 95th percentile is 113.4 kPa in Case 5 (Table 8). These are modeled loading patterns, obtained with a uniform 2 mm wall, literature material parameters and fixed ends. The retained records do not include the coupling-step settings, and local tissue strength is unknown, so the stress fields do not identify a failure site or quantify rupture risk.

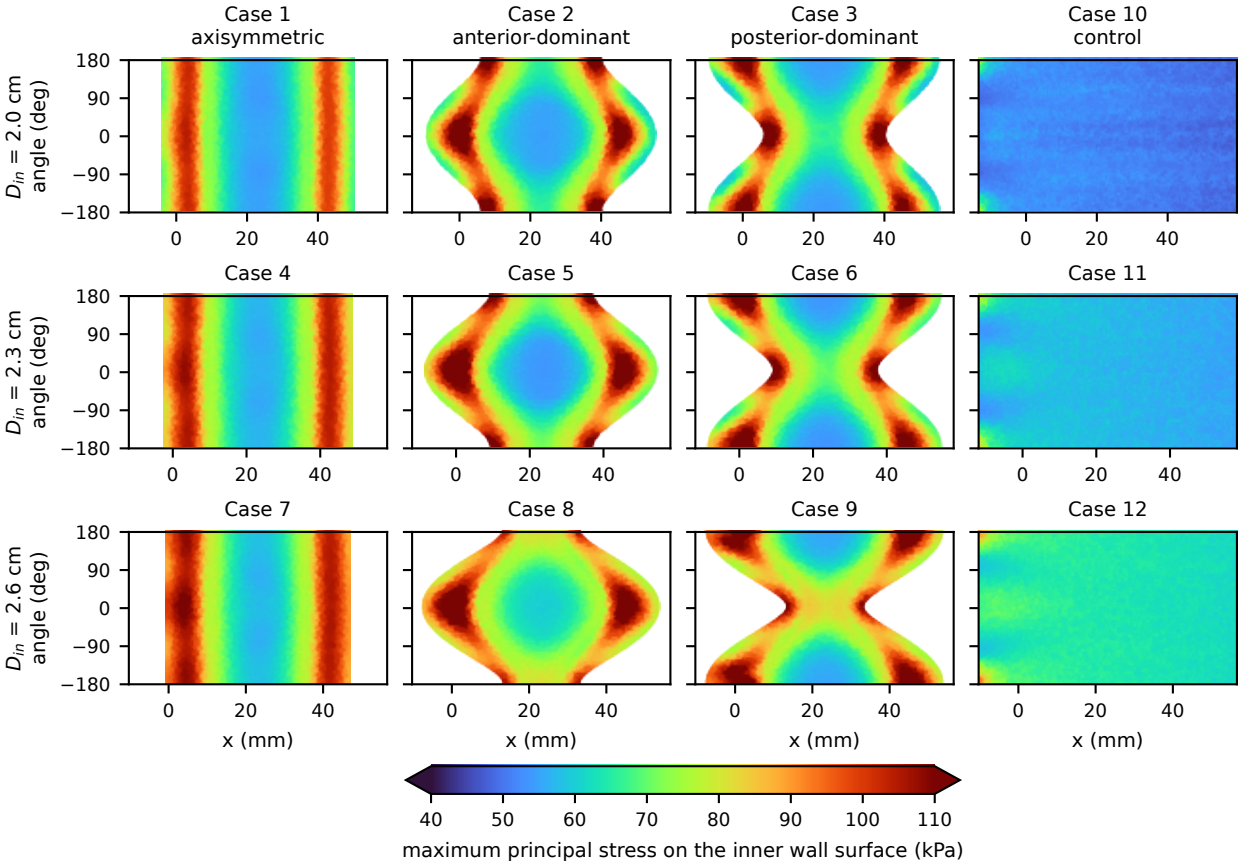


Figure 13: MPS on the inner wall surface of the dilation segment, unwrapped as in Figure 9, on one color scale. Rows are inlet diameters and the last column is the matching control.

Table 8: Maximum principal stress (kPa) on the inner wall surface of the dilation segment. Sac-body medians use the middle third of the segment, within 60° of the anterior or posterior direction; the neck value is the 95th percentile over the outer thirds.

			Segment		Sac body, median		Necks
Case	D_{in} (cm)	Morphology	Median	95th pct.	Anterior	Posterior	95th pct.
1	2.0	axisymmetric	67.6	99.5	56.7	56.7	101.0
2	2.0	anterior-dominant	70.2	106.9	55.9	69.2	111.3
3	2.0	posterior-dominant	69.8	105.5	68.3	55.8	110.0
10	2.0	control	52.5	54.8	52.4	52.8	54.9
4	2.3	axisymmetric	71.4	103.9	58.1	59.0	104.5
5	2.3	anterior-dominant	72.4	108.5	55.0	73.6	113.4
6	2.3	posterior-dominant	72.2	107.4	72.9	55.3	112.1
11	2.3	control	58.4	60.7	58.7	58.9	61.0
7	2.6	axisymmetric	74.1	105.2	59.4	60.6	106.3
8	2.6	anterior-dominant	74.9	106.8	61.8	78.1	111.8
9	2.6	posterior-dominant	76.5	107.3	80.3	57.2	112.9
12	2.6	control	65.1	67.6	65.3	65.7	68.0

4.6 Plane-wise flow summaries

Plane-wise speed and TKE summaries describe how the flow changes along the vessel and where the sac-associated disturbance persists downstream.

The eight planes are shown in Figure 1, and their diameters are listed in Table 3. Figure 14 shows the retained plane summaries for every case; Table 9 gives group medians. Speed is higher at D7–D8 than at D1–D3, consistent with the narrower distal lumen [55]. In the aneurysmal cases it falls to 0.10–0.16 m s^{-1} at D5, where every sac reaches 4.0 cm, compared with a control median of 0.193 m s^{-1}, and recovers by D7. Continuity constrains cross-sectional mean axial velocity through flow rate and area; the plotted mean speed also reflects recirculation and the averaging used in the retained summaries. These prescribed-flow models do not test downstream perfusion.

Modeled k rises through the sac, reaches about twice the control value at D5, remains elevated at D6–D7 and returns toward the control level at D8. The asymmetric sacs have higher D5 values than the axisymmetric sacs at each inlet diameter. These profiles are consistent with the sac recirculation and distal reattachment seen in the streamline maps [56]. They locate the modeled disturbance but do not directly resolve the underlying turbulent fluctuations.

The plane summaries place the largest speed reduction at the sac center and the elevated modeled TKE across the sac and its downstream region. They complement the wall quantities by describing the flow within the lumen. Because the slice workbook does not retain its averaging interval, these profiles are treated as descriptive summaries rather than independently verified cycle averages (Section 6.4).

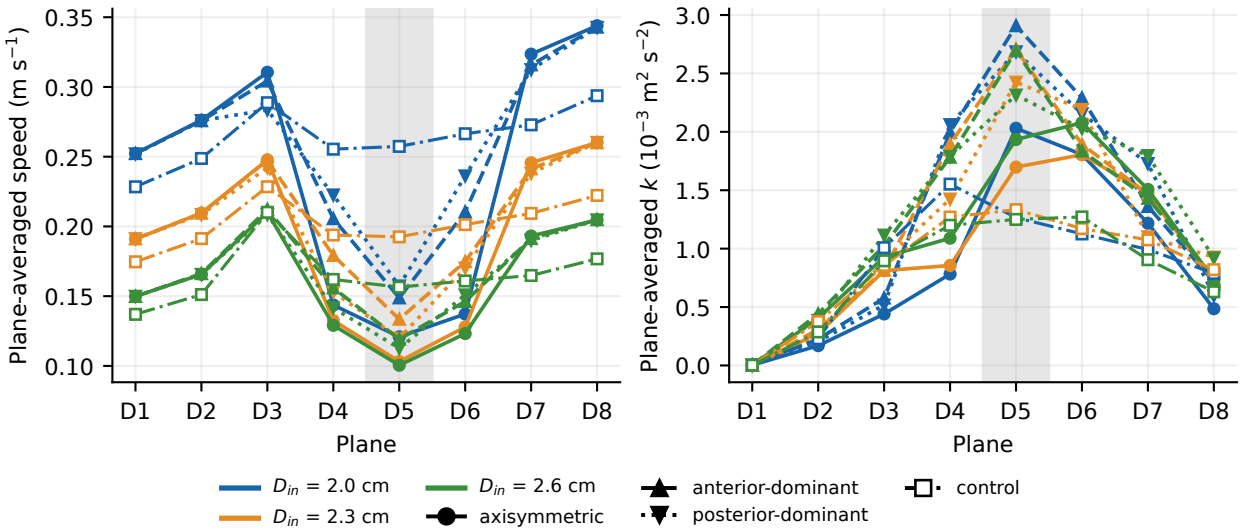


Figure 14: Plane-averaged speed (left) and TKE per unit mass (right) at D1–D8 for all twelve geometries. Color gives the inlet diameter, and line style and marker the morphology; controls are dash-dot with open squares. The band marks the sac center (D5).

Table 9: Descriptive medians of the reported plane-averaged CFD quantities across nine aneurysmal and three control geometries. k is turbulent kinetic energy per unit mass (m^2 s^{-2}). These designed cases do not support patient-population significance tests.

	Mean speed (m s^{-1})		k (10^{-3} m^2 s^{-2})	
Plane	Aneurysmal	Control	Aneurysmal	Control
D1	0.191	0.175	0.002	0.003
D2	0.209	0.191	0.275	0.287
D3	0.247	0.228	0.815	0.896
D4	0.153	0.194	1.783	1.270
D5	0.120	0.193	2.425	1.266
D6	0.151	0.201	2.028	1.171
D7	0.241	0.209	1.468	0.993
D8	0.260	0.222	0.728	0.788

5 Reconstruction results

5.1 Checks on the complete CFD fields

Before any fitting, the complete CFD fields of the nine aneurysmal cases were used to check Eq. (3.2) and to measure the post-processing floors. With 40-neighbor least-squares gradients, the unsteady form with the CFD eddy viscosity and without a $2\rho k/3$ term gave the smallest momentum residual in every case and phase, at 0.03–0.17 of the convective term in the bulk of the lumen. Omitting the time derivative raised the residual to 0.18–0.30 of convection at systole, and to 0.45–0.56 inside the sac, and to 0.49–0.73 at diastole, and adding the $2\rho k/3$ term made the residual larger. A quasi-steady momentum residual is therefore not consistent with these transient fields, particularly inside the sac.

Table 10 gives the floors obtained by passing the complete CFD velocity through the

shared post-processing. Integrating the CFD's own pressure gradient reproduces the CFD pressure to 0.5–0.9%, so the integration itself is accurate, and the remaining floor comes from the momentum terms. With the unsteady form and oracle μ_t the pressure floor is 1.6–2.3% at systole, except for Case 2, whose forward window gives 7.1%, and 2.2–4.1% at diastole. Information matters differently in the two phases. At systole the molecular form is almost as good as the oracle for the 2.3 and 2.6 cm cases (1.8–2.7%) but not for the 2.0 cm cases (4.9–11.3%), in line with their larger eddy viscosity (Table 6). At diastole removing either the time term or the eddy viscosity degrades the floor substantially, to 3.9–10.3% and 9.0–19.3%, respectively. The WSS floor depends strongly on the distance h in Eq. (3.4). At systole it is 0.07–0.09 at $h = 0.1$ mm but 0.14–0.27 at the prespecified $h = 0.25$ mm, because the systolic velocity profile is no longer close to quadratic over the $2h = 0.5$ mm stencil, whereas at diastole it is 0.04–0.06 at either distance.

Table 10: Pipeline floor, obtained by passing the complete CFD velocity through the shared post-processing. Pressure columns give the sub-cloud relative L2 error for each information setting (time term on or off; CFD eddy viscosity or molecular viscosity only). Δp_{sub} is the CFD pressure drop between the inlet and outlet ends of the sub-cloud. Wall-shear columns give the aneurysm-zone magnitude error of the one-sided estimate at $h = 0.1$ mm and at the registered $h = 0.25$ mm. Case 2's systolic window runs forward from the target (1.780–1.788 s), which explains its larger unsteady error.

		Pressure relative L2					Wall shear	
Case	Phase	Unst.+μ_t	Steady+μ_t	Unst.+μ	Steady+μ	Δp_{sub} (Pa)	$h = 0.1$	$h = 0.25$
1	Sys.	0.022	0.031	0.049	0.067	986.1	0.086	0.249
1	Dia.	0.035	0.074	0.152	0.188	8.8	0.044	0.042
2	Sys.	0.071	0.027	0.113	0.048	1063.4	0.090	0.273
2	Dia.	0.037	0.056	0.145	0.154	1.1	0.048	0.042
3	Sys.	0.022	0.029	0.065	0.071	1010.2	0.090	0.220
3	Dia.	0.024	0.039	0.091	0.101	3.7	0.048	0.046
4	Sys.	0.019	0.037	0.023	0.036	664.2	0.077	0.182
4	Dia.	0.032	0.103	0.175	0.242	5.5	0.050	0.037
5	Sys.	0.019	0.036	0.027	0.036	682.4	0.094	0.208
5	Dia.	0.031	0.051	0.114	0.121	0.4	0.052	0.041
6	Sys.	0.018	0.033	0.026	0.032	685.4	0.078	0.191
6	Dia.	0.041	0.068	0.193	0.211	5.8	0.054	0.042
7	Sys.	0.016	0.037	0.018	0.035	412.8	0.070	0.151
7	Dia.	0.038	0.090	0.172	0.246	4.1	0.050	0.040
8	Sys.	0.023	0.048	0.018	0.042	422.4	0.090	0.183
8	Dia.	0.039	0.065	0.154	0.160	0.6	0.055	0.043
9	Sys.	0.020	0.033	0.020	0.033	397.3	0.070	0.141
9	Dia.	0.022	0.045	0.090	0.096	4.8	0.060	0.048

5.2 Momentum-constrained PINNs on the development geometries

In the development cases, every neural field reconstructed the hidden velocity three to four times more accurately than RBF (relative errors 0.037–0.052 versus 0.13–0.16 at systole)

and five times more accurately than linear interpolation. However, the prespecified criterion of the development test failed on both endpoints (Table 11). The momentum arms were trained with a blood density of 1060 kg m^{-3} in the scaling of their residual before the density was corrected to 1050 kg m^{-3}; all reported scores use the corrected value, The difference is about 1%, but its effect on optimization was not tested by retraining. The gate scored the momentum PINNs by their own pressure output and automatic-differentiation WSS, the direct route that a PINN offers, and the other methods by shared post-processing, so the comparison changes both the fitting and the pressure route. The unsteady PINN's own pressure output had errors of 0.43–0.53 at systole and 0.56–1.05 at diastole, and the steady PINN's 0.51–0.90, whereas the continuity field passed through the shared integration reached 0.023–0.049. The PINN's automatic-differentiation WSS (0.40–0.45 at systole) was also worse than the pipeline WSS of the continuity field (0.35–0.37).

Because the gate changed both the fitting and the pressure route, we also passed the velocity of every momentum PINN through the shared post-processing (Table 12). In the prespecified unsteady form the unsteady PINN's velocity still gave pressure errors of 0.37–0.48 at systole and 0.58–1.11 at diastole, against 0.023–0.049 for the continuity field, because that form uses the difference between the field's two snapshots. Dropping the time term reduced the error to 0.037–0.089, still above the continuity field. The PINN's momentum residual was an order of magnitude smaller than that of the continuity field (0.2 against 2.5 in scaled units), so the network satisfied the momentum balance by adjusting a time derivative that was constrained only at the observed nodes of two snapshots 5–10 ms apart. With so little temporal information, $\partial_t \mathbf{u}$ and the pressure act as slack variables in a soft momentum penalty rather than as information. The steady PINN, whose loss does not contain a time derivative, came close to the continuity field only in the systolic window of Case 1 (0.029 against 0.025). For WSS the shared estimate gave similar errors for all fields at systole (0.34–0.40), and the momentum PINNs gave lower errors in the diastolic Case 9 window (0.11–0.12 against 0.22). Adding the momentum balance to the loss therefore did not improve the velocity used for pressure, although it helped WSS in one diastolic window. The confirmation study therefore evaluated the continuity field with physics applied after fitting, as specified in Section 3.8.

Table 11: Pilot endpoints. The continuity field is scored through the shared post-processing, and the momentum PINNs through their own pressure output and automatic-differentiation WSS. Values average two draws at systole and one at diastole.

Case	Phase	Cont. pressure (post-proc.)	Steady head (direct)	Unsteady head (direct)	Cont. shear (post-proc.)	Unsteady shear (autograd)
1	Sys.	0.025	0.787	0.435	0.369	0.447
1	Dia.	0.045	0.897	1.048	0.113	0.130
9	Sys.	0.023	0.633	0.531	0.346	0.396
9	Dia.	0.049	0.515	0.558	0.224	0.155

5.3 Confirmation on seven held-out geometries

At the primary 2.5 mm spacing the continuity field followed by the shared post-processing had lower pressure error than RBF with the same post-processing in all 14 case–phase units,

Table 12: Pilot fields scored through the same post-processing. Pressure is the relative L2 error of momentum integration applied to each field's velocity, in the prespecified unsteady form with CFD eddy viscosity, and, for the unsteady PINN, also in the steady form, which does not use the field's time derivative. WSS is the finite-distance estimate at $h = 0.25$ mm. All fields use the CFD eddy viscosity. Values are averaged over two draws at systole and one at diastole.

		Pressure				Aneurysm-zone WSS		
Case	Phase	Cont. field	Steady PINN	Unsteady PINN	Unsteady PINN, steady form	Cont. field	Steady PINN	Unsteady PINN
1	Sys.	0.025	0.029	0.373	0.046	0.369	0.399	0.401
1	Dia.	0.045	0.060	1.103	0.089	0.113	0.118	0.100
9	Sys.	0.023	0.036	0.479	0.037	0.346	0.347	0.335
9	Dia.	0.049	0.126	0.577	0.050	0.224	0.123	0.111

with a median ratio of 0.458, and lower aneurysm-zone WSS error in all 14 units, with a median ratio of 0.508 (Table 13, Figure 15). Both prespecified hypotheses were met, and the velocity guardrail held in every unit, with hidden-velocity errors of 0.033–0.050 against 0.113–0.169 for RBF. At the level of individual draws the WSS advantage held in 42 of 42 comparisons, whereas the pressure advantage held in 39 of 42. The prespecified decision rule was the count of units, and no significance test was planned. The draw-level counts are descriptive because draws and phases share geometries; they are not treated as independent replicates for significance testing. The three adverse pressure draws were all diastolic, in Case 4 (ratio 1.05), Case 7 (1.40) and Case 8 (1.75), and the case-wise tables in Appendix A show that the diastolic pressure error of the neural field varies between draws by up to 0.085, several times more than RBF's.

Table 13: Median of seven case means at 2.5 mm; each reconstructed case mean uses three paired draws. Entries are node-weighted relative L2 errors. Pressure is conditional on oracle CFD eddy viscosity. Full CFD denotes the same post-processing applied to complete velocity; its error is a reference, not a strict lower bound.

Phase	Quantity	Continuity	Data-only	RBF	Linear	Full CFD
Systole	Velocity	0.044	0.052	0.146	0.226	0.000
Systole	Pressure	0.023	0.028	0.079	0.292	0.019
Systole	Wall shear	0.356	0.333	0.609	0.834	0.191
Diastole	Velocity	0.038	0.039	0.130	0.242	0.000
Diastole	Pressure	0.094	0.063	0.132	0.167	0.037
Diastole	Wall shear	0.163	0.161	0.374	0.615	0.042

The phases differ in how close the reconstruction comes to its floor. At systole the median pressure error of the continuity field, 0.023, is close to the floor of 0.019, so the reconstructed velocity is good enough that the post-processing, not the reconstruction, sets most of the pressure error. At diastole the median error of 0.094 is two and a half times the floor of 0.037, and the advantage over RBF (0.132) is small in Cases 7 and 8 (0.155 against 0.171 and 0.101 against 0.104). The median systolic WSS error of 0.356 lies well above its floor of 0.191, and the diastolic 0.163 lies well above 0.042, so for WSS the reconstruction remains the limiting step.

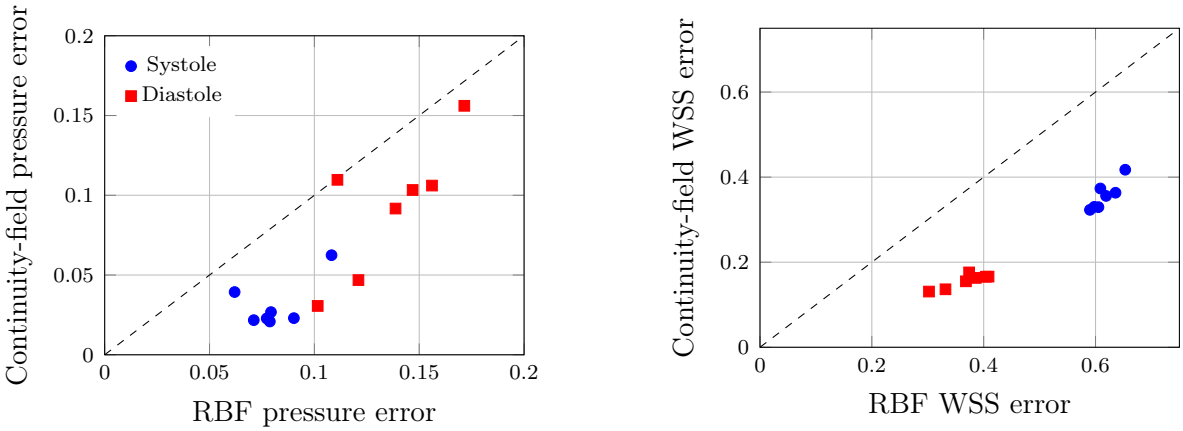


Figure 15: Case-wise confirmation errors for pressure (left, unsteady form with oracle μ_t) and aneurysm-zone WSS magnitude (right, $h = 0.25$ mm) at 2.5 mm. Each point is one case–phase unit averaged over three paired draws; points below the diagonal favor the continuity field. Both pipelines receive the same oracle eddy viscosity.

5.4 Local errors and field maps

Global relative errors can hide local failures, so Table 14 and Figures 16–18 report local measures. The 95th percentile of the pointwise velocity error magnitude in systole is 0.057 m s^{-1} for the continuity field and 0.20 m s^{-1} for RBF, against peak speeds of about 1.0–1.6 m s^{-1}, and within the sac the relative velocity error is 0.075 against 0.185. The largest pointwise velocity errors of the continuity field lie in the thin shear layer where the flow separates at the bend exit (Figure 16).

Table 14: Additional error measures at 2.5 mm, reported as medians of seven case means. Velocity p95 is the 95th percentile of pointwise vector-error magnitude; sac velocity is relative L2. The shear ratio compares predicted and CFD 99th percentiles. Pressure-drop error is averaged in absolute value within each case before taking the median.

Phase	Method	Velocity p95 (m/s)	Sac velocity rel. L2	Shear q99 ratio	Drop error (Pa)
Systole	Continuity	0.0569	0.075	0.923	5.57
Systole	Data-only	0.0657	0.082	0.956	8.64
Systole	RBF	0.2005	0.185	0.520	30.59
Diastole	Continuity	0.0056	0.040	0.897	0.48
Diastole	Data-only	0.0057	0.041	1.001	0.47
Diastole	RBF	0.0206	0.143	0.611	0.67

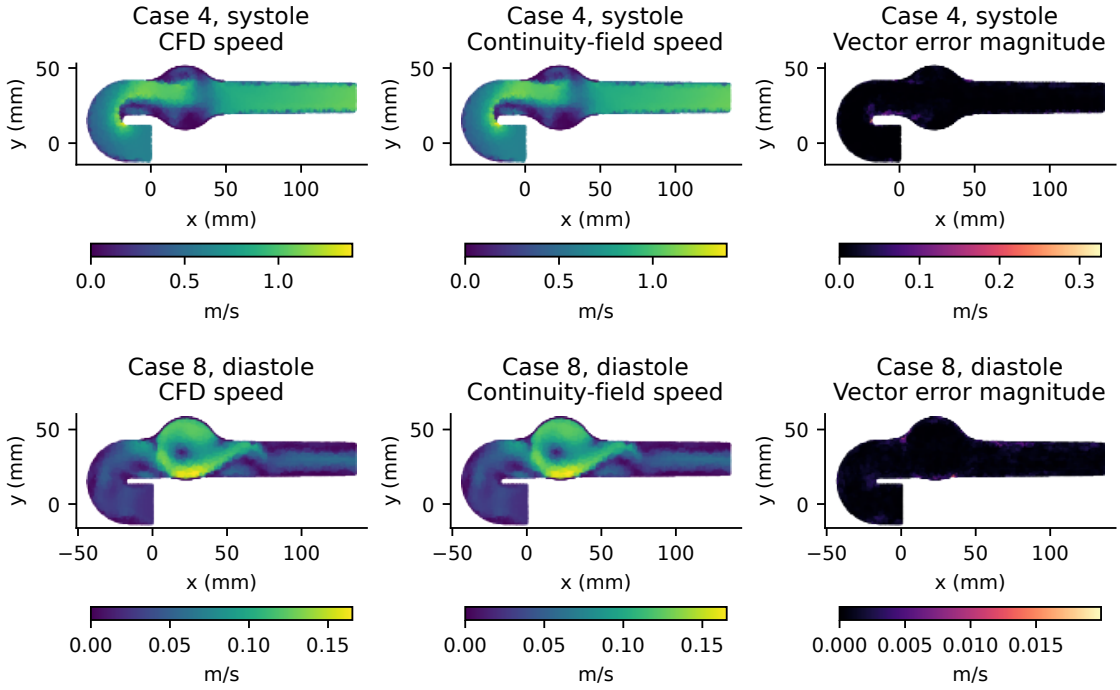


Figure 16: Hidden-node speed on the mid-plane ($|z - z_{\mathrm{mid}}| < 0.35$ mm) for Case 4 at systole and Case 8 at diastole (draw 0, 2.5 mm). Left, CFD; center, continuity field; right, magnitude of the vector error. Only unobserved nodes are shown.

The 99th percentile of aneurysm-zone WSS is recovered to 0.92 of the CFD value at systole and 0.90 at diastole, whereas RBF recovers only 0.52 and 0.61 of it, so RBF's WSS error consists mostly of flattened peaks. The Pearson correlation between reconstructed and CFD WSS magnitude is 0.92 at systole and 0.94 at diastole for the continuity field, against 0.87 and 0.79 for RBF, and the WSS vector errors are 0.38 and 0.20 against 0.62 and 0.42. The systolic pressure drop, which ranges from 413 to 1063 Pa across the cases, is recovered with a median absolute error of 6 Pa against 31 Pa for RBF. At diastole the median drop errors of 0.48 and 0.67 Pa are comparable to the drops themselves (0.4–5.8 Pa), so neither method recovers the diastolic drop reliably, even though the relative error over the whole pressure field is modest.

The two kinds of error also differ in form. In the median systolic unit (Figure 17) the continuity field reproduces the WSS rings at both necks, while RBF smears them and underestimates their peaks.

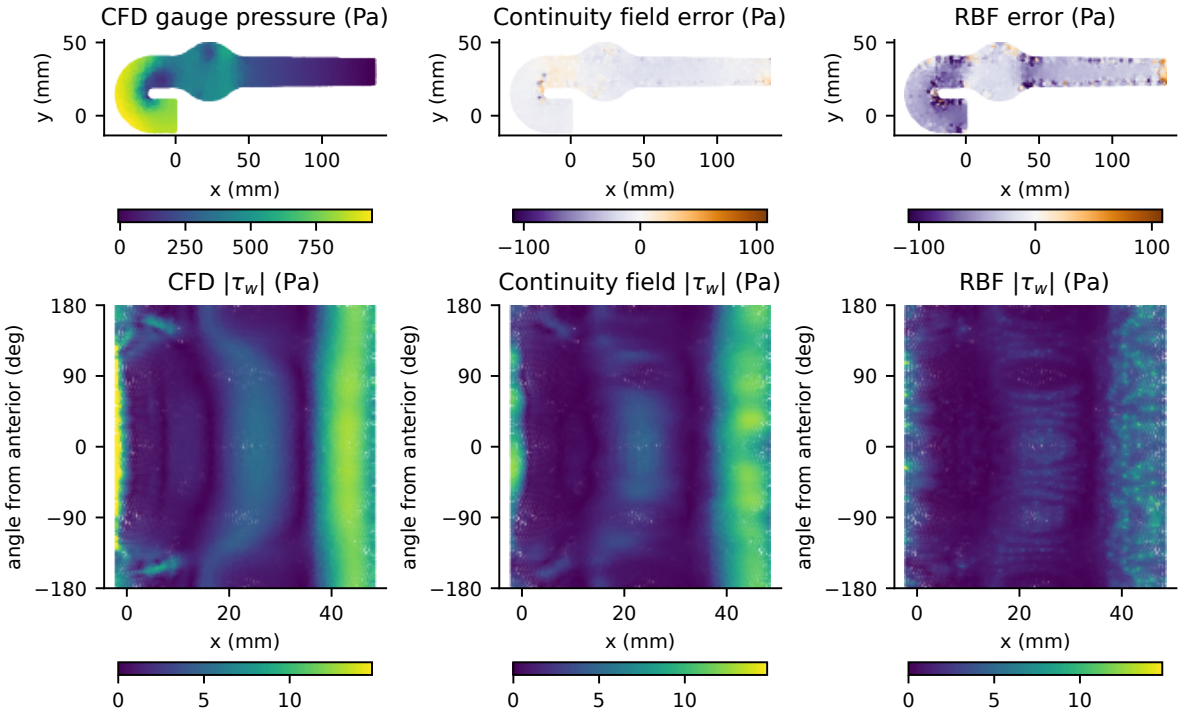


Figure 17: Case 4 at systole (draw 0, 2.5 mm), close to the median result. Top, CFD mid-plane pressure and the pressure errors of the continuity field and RBF. Bottom, aneurysm-zone WSS unwrapped about the vessel axis. Relative errors are 0.024 and 0.100 for pressure and 0.32 and 0.60 for WSS.

In the adverse diastolic unit (Figure 18), the continuity field's pressure error is a smooth offset of about 1 Pa that accumulates along the vessel, which points to a small bias in the integrated pressure gradient such as the two-snapshot acceleration, while RBF's pressure errors are concentrated near the sac necks.

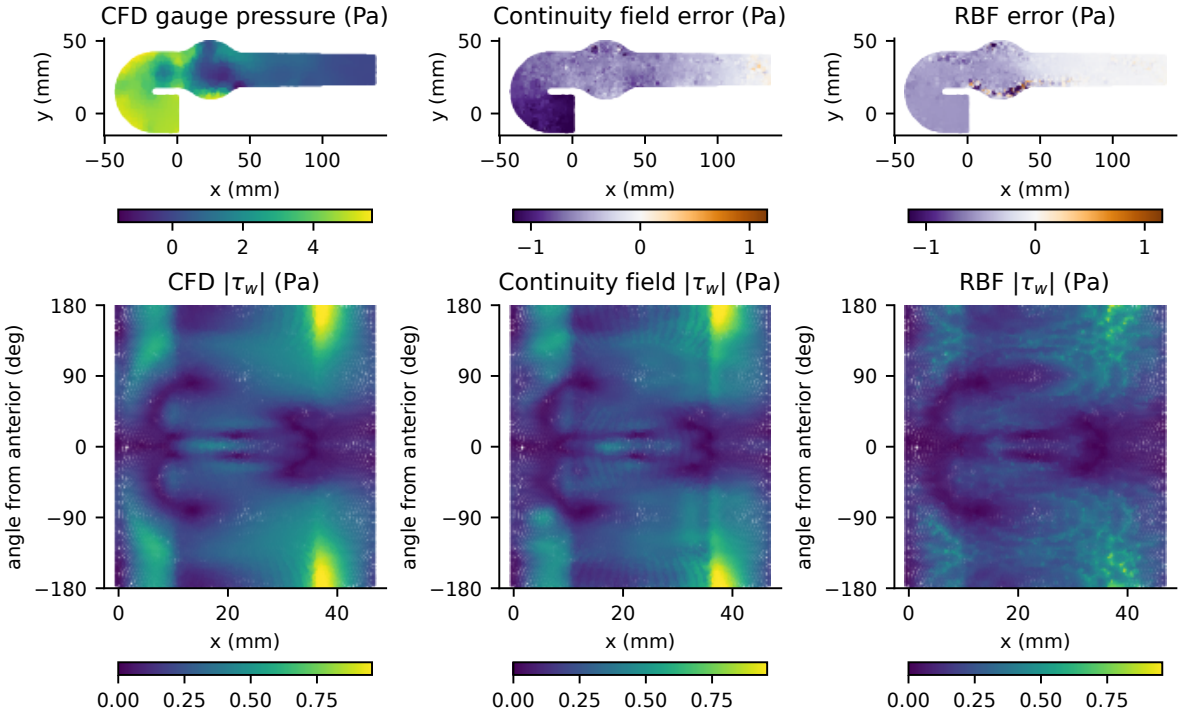


Figure 18: As Figure 17, for the adverse unit Case 7 at diastole. Relative errors are 0.25 and 0.18 for pressure and 0.14 and 0.33 for WSS. The CFD pressure varies by only a few pascals, so errors of about 1 Pa dominate.

The WSS calculated by automatic differentiation of the continuity field itself was worse than the finite-distance estimate from the same field (median 0.55 against 0.36 in systole and 0.27 against 0.16 in diastole). The soft no-slip penalty fixes the velocity at the wall but leaves its wall-normal gradient poorly determined, so the network's analytic derivative should not be read as WSS. The finite-distance estimate is also sensitive to h. Evaluated at 0.1 mm instead of 0.25 mm, the continuity field's median systolic WSS error falls from 0.356 to 0.326, and at 0.5 mm it rises to 0.476. We keep the prespecified distance because choosing h with knowledge of the WSS target would be information that a deployment does not have.

5.5 Contribution of the continuity penalty and sampling density

Figure 19 separates the effect of the continuity penalty from that of the neural representation. At 2.5 mm, the data field without an interior PDE penalty is close to the continuity field, with median velocity errors of 0.052 and 0.039 versus 0.044 and 0.038, and is slightly better on two endpoints, diastolic pressure (0.063 versus 0.094) and systolic WSS (0.333 versus 0.356). At the densest spacing of 1.5 mm the data field is better on every endpoint except systolic pressure. Only at the sparsest spacing of 4.0 mm, where the observations cover 0.34–0.37% of the nodes, does the continuity penalty help consistently, reducing the median

hidden-velocity error from 0.125 to 0.077 at systole and from 0.105 to 0.077 at diastole, the systolic pressure error from 0.072 to 0.052 and the diastolic WSS error from 0.371 to 0.316. The main advantage over RBF therefore comes from fitting one smooth space-time representation to both snapshots, and continuity becomes useful when the data alone no longer determine the field.

The advantage of both neural fields over RBF grows as the observations become sparser. The median systolic pressure errors at 1.5, 2.5 and 4.0 mm are 0.024, 0.023 and 0.052 for the continuity field versus 0.029, 0.079 and 0.178 for RBF, and the median systolic WSS errors are 0.30, 0.36 and 0.48 versus 0.49, 0.61 and 0.68. At diastole and 4.0 mm the continuity field's pressure error is 0.096 versus 0.318 for RBF. Only one draw was performed at 1.5 and 4.0 mm, so these values describe a trend rather than an uncertainty interval.

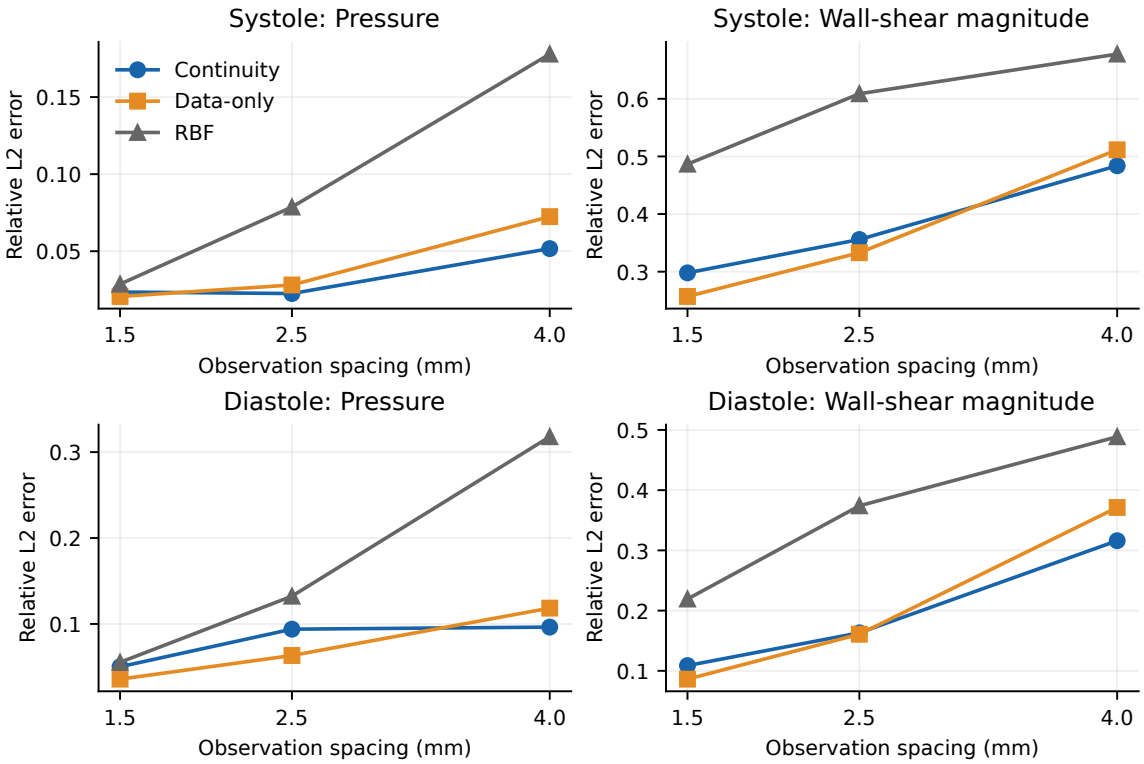


Figure 19: Median pressure (unsteady form with oracle μ_t) and aneurysm-zone WSS errors over Cases 2–8 against observation spacing. The 2.5 mm values average three draws per case; 1.5 and 4.0 mm use one draw.

Figure 20 asks two separate questions about pressure recovery, namely whether to retain the local acceleration term and whether to supply the CFD eddy viscosity. Here, steady and unsteady refer to the pressure calculation applied after reconstruction, not to separately trained neural fields. In the seven confirmation geometries, the systolic median improves when the time term is retained and changes little when the eddy viscosity is removed. The unsteady form gives a median error of 0.023 against 0.034 for the steady form, close to the floors of 0.019 and 0.036, and dropping μ_t changes 0.023 to 0.027. At diastole the eddy viscosity is essential, since removing it more than doubles the continuity field's error, from 0.094 to 0.207, and raises the floor from 0.037 to 0.154. The time term behaves differently for the reconstruction and for the complete CFD field at diastole. The floor favors the

unsteady form (0.037 against 0.065), but the continuity field does better with the steady form (0.072 against 0.094), because the 10 ms finite difference of two reconstructed diastolic fields, whose speeds are about one tenth of the systolic ones, amplifies small reconstruction errors. Diastolic pressure recovery in this setting therefore needs both an estimate of the eddy viscosity and a more accurate acceleration than two sparse snapshots provide.

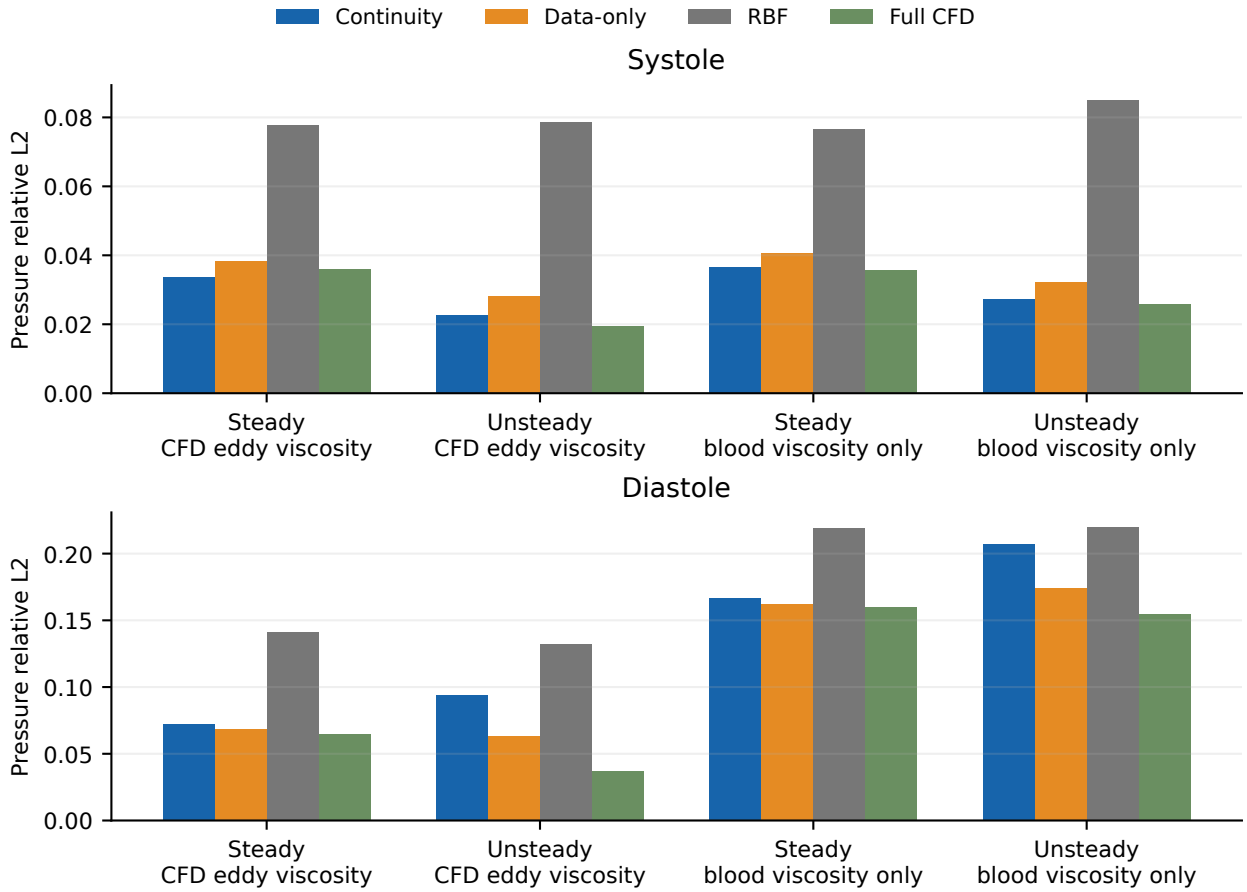


Figure 20: Median pressure error over Cases 2–8 at 2.5 mm for the four post-processing settings. Unsteady retains the two-snapshot time derivative and steady omits it. "CFD eddy viscosity" uses $\mu + \mu_t$ from the reference simulation, and "blood viscosity only" sets $\mu_t = 0$. Full CFD applies the same settings to the complete velocity.

5.6 Robustness to the choice of interpolation comparator

The prespecified RBF used one fixed local interpolant in each snapshot. To test how much the comparison depended on that choice, we also evaluated a tuned spatial RBF and a space-time RBF, with their settings chosen from the withheld velocity observations. Table 15 reports the completed comparisons for draw 0 at 2.5 mm; these values use one draw per case, so they should be compared within this table rather than directly with the three-draw means of the primary analysis.

At systole, fitting both snapshots with a space-time RBF changed the median pressure error only slightly, from 0.079 for the prespecified RBF to 0.074, and the WSS error from 0.606 to 0.595. The continuity field gave errors of 0.025 and 0.356, respectively, and had

lower velocity, pressure and WSS errors than each RBF variant in all seven cases. Tuning the spatial RBF did not improve this phase, since its median pressure and WSS errors were 0.127 and 0.686; therefore, choosing an interpolant by its velocity error in withheld observations did not ensure lower errors in hidden fields derived from its spatial gradients.

At diastole, both alternatives helped the interpolation. Relative to the prespecified RBF, the tuned spatial RBF reduced the median velocity error from 0.129 to 0.100, the pressure error from 0.136 to 0.111 and the WSS error from 0.371 to 0.321, and the space-time RBF reduced them to 0.096, 0.113 and 0.301. The continuity field kept lower velocity and WSS errors than both variants in all seven cases, with a median pressure error of 0.076, but its pressure advantage held in only five of the seven diastolic cases against each variant. Over all 14 case–phase units, the continuity field had the lower pressure error in 12 against the tuned RBF and in 12 against the space-time RBF, with median error ratios of 0.50 and 0.54, and the lower WSS error in all 14 against both, with median ratios of 0.54 and 0.58. It also had a lower pressure error than the better of the two variants in 12 of the 14 units. The advantage persists against the two additional interpolants tested, although they narrow the diastolic pressure gap. This does not exclude other interpolation schemes from performing better.

Table 15: Robustness to the interpolation comparator, given as median relative errors over the 14 case–phase units of Cases 2–8 with draw 0 at 2.5 mm. The tuned RBF chooses kernel, neighbourhood and smoothing on the withheld observations; the space-time RBF interpolates both snapshots jointly with a time scale chosen the same way. All methods share the post-processing; pressure uses the unsteady form with oracle μ_t.

Phase	Method	Velocity	Pressure	WSS
Systole	Continuity field	0.043	0.025	0.356
Systole	RBF (prespecified)	0.144	0.079	0.606
Systole	RBF (tuned)	0.177	0.127	0.686
Systole	RBF (space-time)	0.147	0.074	0.595
Diastole	Continuity field	0.038	0.076	0.173
Diastole	RBF (prespecified)	0.129	0.136	0.371
Diastole	RBF (tuned)	0.100	0.111	0.321
Diastole	RBF (space-time)	0.096	0.113	0.301

5.7 Sensitivity to near-wall observations

Because the grid observations are snapped to mesh nodes, some lie on the near-wall prism layers (Section 3.3). To measure how much the results depend on them, we repeated draw 0 at 2.5 mm for the continuity field and RBF after removing every observation closer than $\Delta/2 = 1.25$ mm to the wall, which removed 28–32% of the observations in each case (Table 16). Nothing else was changed.

The WSS errors rose sharply. For the continuity field the median rose from 0.356 to 0.806 at systole and from 0.173 to 0.521 at diastole, and for RBF from 0.606 to 0.886 and from 0.371 to 0.681. The accuracy of the one-sided WSS estimate therefore rests largely on velocity sampled within about a millimetre of the wall, which a 2.5 mm imaging voxel would not resolve. Pressure was much less sensitive. The continuity field's median pressure error

changed from 0.025 to 0.027 at systole and from 0.076 to 0.045 at diastole, consistent with pressure being integrated only on nodes more than 1 mm from the wall. The continuity field still had lower velocity, pressure and WSS errors than RBF in all 14 units, so the comparison between the methods does not depend on the near-wall samples, whereas the absolute WSS accuracy does.

Table 16: Sensitivity to near-wall observations. Median relative errors over Cases 2–8 for draw 0 at $\Delta = 2.5$ mm, with all observations and with the observations closer than $\Delta/2 = 1.25$ mm to the wall removed (28–32% of them). Pressure uses the unsteady form with CFD eddy viscosity; WSS uses $h = 0.25$ mm. Velocity is scored on all unobserved nodes, so the second pair of rows is also scored on the removed nodes.

Phase	Method and observations	Velocity	Pressure	WSS
Systole	Continuity field, all observations	0.043	0.025	0.356
Systole	RBF, all observations	0.144	0.079	0.606
Systole	Continuity field, none within 1.25 mm	0.130	0.027	0.806
Systole	RBF, none within 1.25 mm	0.248	0.121	0.886
Diastole	Continuity field, all observations	0.038	0.076	0.173
Diastole	RBF, all observations	0.129	0.136	0.371
Diastole	Continuity field, none within 1.25 mm	0.123	0.045	0.521
Diastole	RBF, none within 1.25 mm	0.258	0.108	0.681

6 Discussion

6.1 Effects of inlet diameter and asymmetry

With the flow waveform and the maximum sac diameter held fixed, inlet diameter set the strength of the systolic disturbance and asymmetry set its location. A narrower parent vessel raised the peak Reynolds number, the jet speed entering the sac, the pressure drop, the fraction of reversed flow and the modeled eddy viscosity, and it strengthened the high-shear rings at the necks. This agrees with reports that a larger parent vessel lowers WSS in intracranial aneurysms [15], and it extends the comparison to transitional flow in a curved, tapered thoracic segment. Because the same flow rate was prescribed for all diameters, the diameter effect here combines the change in geometry with the change in mean velocity. A study that scaled the flow rate with the vessel area would separate these two influences.

Asymmetry moved the low-shear region and the dominant vortex to the side of the larger bulge while changing the systolic pressure drop by less than 7%. At diastole its effect became larger than that of diameter, with the asymmetric sacs carrying about two to three times the WSS of the axisymmetric ones. A previous study of idealized abdominal and descending thoracic aneurysms also found that asymmetry mainly redistributes recirculation and WSS [45]. The present results add that the redistribution depends on the phase of the cycle, so single-instant comparisons at peak systole understate it. None of these comparisons is a population statistic, since each geometry is a single designed case, and whether low or high WSS relates to growth or rupture cannot be settled by simulations of this kind [1, 8].

6.2 Why the neural field outperformed interpolation

The clearest gain came from the reconstructed velocity rather than from adding momentum to the training loss. The neural fields and the prespecified RBF received the same velocity observations and boundary values, and their pressure and WSS were computed by the same post-processing. Yet the data field, which has no interior PDE penalty, performed close to the continuity field at 1.5 and 2.5 mm. Both neural fields also retained near-wall gradients that the prespecified RBF tended to smooth out, and their finite-distance shear estimates recovered about 90–100% of the CFD 99th-percentile WSS, compared with about half for RBF at systole. The continuity penalty became consistently useful at the sparsest spacing, 4.0 mm.

The network fits one function across the domain and both snapshots, whereas the prespecified RBF uses a local neighborhood at each time. This difference offers a plausible explanation for the results, but the study does not isolate the effects of spatial smoothness, joint temporal fitting and the network architecture. The space-time RBF check narrows that interpretation, because fitting both snapshots jointly did not close the gap to the neural field at systole. At diastole, both tuning and joint temporal fitting improved all three median errors of the interpolant and reduced the continuity field's pressure advantage to five of seven cases for draw 0. These checks support the use of a neural representation in this setting without establishing that it is preferable to every interpolation scheme. Related work has used implicit neural representations for 4D flow reconstruction without a physics loss [37]; here, the representation supplies velocity to the established momentum-integration route for relative pressure [28, 29].

The momentum PINNs did not meet the criterion of the development test. Their poor pressure outputs, despite small momentum residuals, are consistent with a weakly determined trade-off between pressure and local acceleration. Two velocity snapshots constrain a finite difference, but they do not uniquely determine the network's time derivative between them. The diagnostic in Section 5.2 showed that a small residual could coexist with an inaccurate pressure field. This fits the broader sensitivity of PINNs to loss weighting and the information available to each residual [40, 41]. Recovering pressure after fitting avoids relying on that pressure head, although the two-snapshot acceleration still carries reconstruction error. Longer time series, hard constraints and other loss weights might change the comparison. They were not tested here, so the negative result applies to the specified formulation and windows rather than to momentum PINNs in general [42].

6.3 Information needed for each hidden field

Pressure and WSS expose different weaknesses in the reconstruction of the sparse velocity. For the seven confirmation geometries, the retention of local acceleration brought the median systolic pressure error close to the complete-CFD post-processing reference. Removing eddy viscosity did little to change that median, although the reference study of all-cases showed greater sensitivity in narrower geometries of 2.0 cm. The limited systolic sensitivity of the median should therefore not be read as evidence that eddy viscosity can always be neglected.

At diastole, omitting the CFD eddy viscosity raised even the complete-velocity pressure error to 9–19%. The reconstruction results with that field supplied describe a conditional

calculation, because the sparse velocity observations do not themselves provide the closure. Small absolute pressure errors also matter more in this phase. None of the tested methods reliably recovered the diastolic pressure drop, even when its relative error across the field appeared modest. An estimate of eddy viscosity and a better-resolved acceleration are, therefore, separate requirements to examine before applying this calculation to measured data.

For aneurysm-zone WSS, near-wall sampling remained the main limitation. Removing the observations within 1.25 mm of the wall more than doubled the continuity field's WSS error at both phases while leaving the ranking against RBF unchanged (Section 5.7). The error depended on the distance used by the one-sided estimate, and the reconstructed fields remained above the complete-CFD reference. Outside the aneurysm zone, the interior node closest to the wall was a median of 0.58 mm from it; even the complete CFD velocity then gave a poor estimate of the exported shear. This is consistent with the documented sensitivity of velocity-derived WSS to spatial resolution [30, 31]. A smooth velocity fit and an analytic derivative do not, by themselves, establish an accurate wall gradient.

6.4 Limitations

The twelve geometries form a designed family, not a patient cohort. They share one maximum aneurysm diameter, taper ratio and inflow waveform, and they contain no branches. The seven confirmation geometries therefore test variation within that family. The rigid-wall flow solution has reported mesh, time-step and cycle checks for aneurysm WSS, but no experimental validation. The structural calculation uses a uniform 2 mm wall, literature material parameters, fixed ends and a quasi-static coupling. Its coupling-step and mesh-update settings were not retained with the results, and no solid-mesh convergence check is available, so the stress fields are used only to describe modeled stress concentrations.

The temporal records also have limits. Each reconstruction uses two snapshots from a short window. Several instants were retained, but no complete cycle-resolved wall-traction series is available. The solver exported TAWSS to the wall nodes, but its cycle-resolved components did not, so the oscillatory indices were not evaluated. The slice workbook does not state its averaging interval; the slice values were identified as cycle averages from the inlet flow rate.

The observed velocities are noise-free CFD values at the mesh nodes. They do not include partial-volume averaging, noise, segmentation error, or velocity aliasing of 4D flow MRI [27], and some lie within a fraction of a millimeter of the wall (Section 3.3). Without observations within 1.25 mm of the wall, the WSS errors were more than twice as large (Section 5.7), so the absolute WSS precision reported here should not be expected from the imaging data. Performance on measured data may therefore be worse. Pressure calculations with CFD-supplied eddy viscosity additionally assume closure information that would have to be estimated in practice. The errors are node-weighted because the exports contain no cell volumes, which gives a greater weight to the refined near-wall mesh.

The network architecture and training settings were fixed on the development cases without a wider hyperparameter search. The tuned and space-time RBF checks were added after the primary verdict and use one draw per case; they cannot replace the three-draw comparison. A divergence-free interpolant, the natural non-neural counterpart of the conti-

nuity field, was not tested. Every method receives observations in the target geometry. The prediction in a geometry with no observations was not tested. Finally, fitting a neural field took about 20 min on a GPU per window, so its accuracy came at a higher computational cost than interpolation.

7 Conclusion

Twelve idealized descending thoracic aortic geometries, comprising nine aneurysmal models with a fixed maximum sac diameter of 4.0 cm and three nonaneurysmal controls, were simulated with transient transitional CFD, and their common-time fields were used both to compare the geometries and to test the recovery of hidden pressure and WSS from sparse velocity. The inlet diameter dominated the systolic response. Reducing it from 2.6 to 2.0 cm raised the pressure drop 2.3–2.5 times, roughly doubled the fraction of reversed flow in dilation, and raised the 95th-percentile WSS in the necks from 7–8 to 15–18 Pa, while the bulge pattern changed the drop in systolic pressure by less than 7%. Instead, asymmetry relocated the low-shear region and the dominant vortex to the larger bulge, and at diastole it raised the median WSS of the dilation segment 1.8–3.1 times relative to the axisymmetric sacs.

With velocity observed at about 1.3% of the interior nodes at two instants, a continuity-constrained space-time neural field followed by momentum integration and a near-wall shear estimate recovered pressure and aneurysm-zone WSS more accurately than the same post-processing applied to the prespecified RBF in all 14 confirmation units after averaging three draws per unit, with median error ratios of 0.46 and 0.51. The pressure advantage was small at diastole. The benefit came mainly from the neural representation of the velocity. A continuity penalty helped in the sparsest sampling, and momentum PINNs did not meet the development-test criterion for joint pressure and shear recovery. Systolic pressure was recovered close to the accuracy of post-processing itself, diastolic pressure errors increased when the CFD-supplied eddy viscosity was omitted, and WSS accuracy depended on the velocity sampled close to the wall, although the continuity field retained its advantage over RBF without those samples. These findings identify what information each hidden field needs in this idealized, noise-free setting. Measured velocity data, estimated closures, and patient-specific anatomies are the next tests, and none of the results constitutes a clinical rupture-risk assessment.

Declarations

Conflicts of interest. The authors have no conflicts to disclose.

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Author contributions. Conceptualization, M.A.U.R. and M.A.O.; CFD and structural modeling, M.A.U.R.; reconstruction methodology, software and formal analysis of the reconstruction study, M.A.O.; validation, M.A.U.R., M.A.O., F.J. and H.T.; data curation,

M.A.U.R. and M.A.O.; writing (original draft), M.A.U.R. and M.A.O.; writing (review and editing), all authors; funding acquisition, F.J. and H.T. All authors have read and agreed to the submitted version.

Data availability statement. The CFD exports analyzed in this study are available from the corresponding author on reasonable request. The reconstruction code and configuration files are available at `https://github.com/michaelajao/IdealAorta-PINN`.

A Case-wise reconstruction results

Tables 17 and 18 list every confirmation unit at 2.5 mm, spread over the three paired draws and the range of draw-wise error ratios to RBF.

Table 17: Case-wise pressure relative L2 at 2.5 mm, given as the mean (sample standard deviation) over three paired observation/initialization draws. The last column is the range of draw-wise continuity/RBF error ratios, and values above one favor RBF. The spread describes these draws, not patient-population uncertainty.

Case	Phase	Continuity	Data-only	RBF	Ratio range
2	Sys.	0.069 (0.004)	0.069 (0.008)	0.111 (0.010)	0.53–0.68
2	Dia.	0.094 (0.026)	0.050 (0.012)	0.138 (0.016)	0.55–0.81
3	Sys.	0.031 (0.004)	0.030 (0.004)	0.078 (0.009)	0.35–0.47
3	Dia.	0.028 (0.004)	0.032 (0.003)	0.095 (0.014)	0.27–0.32
4	Sys.	0.023 (0.001)	0.027 (0.008)	0.093 (0.019)	0.20–0.30
4	Dia.	0.107 (0.032)	0.071 (0.020)	0.157 (0.018)	0.50–1.05
5	Sys.	0.022 (0.002)	0.028 (0.002)	0.079 (0.011)	0.24–0.31
5	Dia.	0.044 (0.017)	0.059 (0.027)	0.113 (0.015)	0.29–0.56
6	Sys.	0.020 (0.000)	0.027 (0.004)	0.081 (0.017)	0.20–0.30
6	Dia.	0.085 (0.029)	0.093 (0.031)	0.132 (0.004)	0.51–0.89
7	Sys.	0.021 (0.004)	0.022 (0.003)	0.074 (0.015)	0.25–0.34
7	Dia.	0.155 (0.085)	0.090 (0.035)	0.171 (0.008)	0.54–1.40
8	Sys.	0.032 (0.005)	0.030 (0.009)	0.063 (0.007)	0.50–0.54
8	Dia.	0.101 (0.076)	0.063 (0.011)	0.104 (0.005)	0.31–1.75

Table 18: Case-wise wall-shear magnitude relative L2 at 2.5 mm, given as the mean (sample standard deviation) over three paired observation/initialization draws. The last column is the range of draw-wise continuity/RBF error ratios, and values above one favor RBF. The spread describes these draws, not patient-population uncertainty.

Case	Phase	Continuity	Data-only	RBF	Ratio range
2	Sys.	0.417 (0.007)	0.386 (0.018)	0.653 (0.004)	0.63–0.65
2	Dia.	0.166 (0.009)	0.169 (0.029)	0.404 (0.002)	0.39–0.43
3	Sys.	0.363 (0.010)	0.333 (0.026)	0.636 (0.008)	0.55–0.60
3	Dia.	0.166 (0.006)	0.148 (0.013)	0.409 (0.005)	0.39–0.43
4	Sys.	0.329 (0.033)	0.306 (0.018)	0.605 (0.001)	0.51–0.61
4	Dia.	0.131 (0.004)	0.150 (0.020)	0.302 (0.002)	0.42–0.44
5	Sys.	0.356 (0.005)	0.359 (0.011)	0.619 (0.010)	0.56–0.59
5	Dia.	0.163 (0.018)	0.174 (0.008)	0.386 (0.001)	0.37–0.45
6	Sys.	0.373 (0.002)	0.352 (0.027)	0.609 (0.005)	0.61–0.62
6	Dia.	0.155 (0.030)	0.161 (0.008)	0.368 (0.003)	0.33–0.50
7	Sys.	0.323 (0.012)	0.300 (0.011)	0.590 (0.005)	0.53–0.57
7	Dia.	0.136 (0.004)	0.146 (0.019)	0.332 (0.005)	0.40–0.43
8	Sys.	0.330 (0.012)	0.327 (0.009)	0.598 (0.005)	0.54–0.58
8	Dia.	0.176 (0.022)	0.196 (0.013)	0.374 (0.005)	0.40–0.51

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